:- Partial Differentiation :-

Let $z = f(x, y)$ be function of two independent variable x and y. If we keep y const. and x varies then z become a function of x only. Then the ordinary derivative of z w.r to x. keeping y as constant is called partial derivative of z w.r to x. and it is denoted by

$$\frac{\partial z}{\partial x}, \quad \frac{\partial f}{\partial x}, \quad f_x \text{ etc.}$$

Then.

$$\frac{\partial z}{\partial x} = \lim_{\substack{\delta x \to 0 \\ y \text{ const.}}} \frac{f(x+\delta x, y) - f(x, y)}{\delta x}$$

Similarly - the partial derivative of z w.r to y keeping x as constant is denoted by

$$\frac{\partial z}{\partial y} \text{ or } \frac{\partial f}{\partial y} \text{ or } f_y.$$

Then,

$$\frac{\partial z}{\partial y} = \lim_{\substack{\delta y \to 0 \\ x \text{ const.}}} \frac{f(x, y+\delta y) - f(x, y)}{\delta y}$$

The process of finding the differential coefficient of z w.r to one variable keeping other variable const. called partial differentiation

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First order partial derivative :

if $z = f(x, y)$ then the first partial differential coefficient of z w.r.to x or y is called first order partial derivative

for eg.: $\frac{\partial z}{\partial x}$, $\frac{\partial z}{\partial y}$

Higher order Partial derivatives : ⇒

partial derivatives of higher order, of a function $u = f(x, y)$ are obtained by partial differentiation of first order partial derivative

Thus if $u = f(x, y)$ then

$$\frac{\partial^2 u}{\partial x^2} = \frac{\partial}{\partial x}\left(\frac{\partial u}{\partial x}\right),$$

$$\frac{\partial^2 u}{\partial y \partial x} = \frac{\partial}{\partial y}\left(\frac{\partial u}{\partial x}\right)$$

$$\frac{\partial^2 u}{\partial y^2} = \frac{\partial}{\partial y}\left(\frac{\partial u}{\partial y}\right)$$

$$\frac{\partial^2 u}{\partial x \partial y} = \frac{\partial}{\partial x}\left(\frac{\partial u}{\partial y}\right)$$

are called second order partial derivative. Similarly, other higher order derivative can also be obtained.

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Note: ① If $u = f(x, y)$ possesses continuous second order partial derivatives $\frac{\partial^2 u}{\partial x \partial y}$ and $\frac{\partial^2 u}{\partial y \partial x}$

then $\frac{\partial^2 u}{\partial x \partial y} = \frac{\partial^2 u}{\partial y \partial x}$. This is called commutative property.

② Standard rule for differentiation of sum, difference product and quotient are also applicable for partial differentiation.

Ex :⇒ If $z(x+y) = x^2+y^2$ then prove that

$$\left(\frac{\partial z}{\partial x} - \frac{\partial z}{\partial y}\right)^2 = 4\left[1 - \frac{\partial z}{\partial x} - \frac{\partial z}{\partial y}\right]$$

Soln :⇒ Given

$$z(x+y) = (x^2+y^2)$$

$$\Rightarrow z = \frac{(x^2+y^2)}{(x+y)} \quad \text{---(1)}$$

Differentiating (1) partially w.r.to. x, we get

$$\frac{\partial z}{\partial x} = \frac{\partial}{\partial x}\left[\frac{(x^2+y^2)}{(x+y)}\right].$$

$$= \frac{(x+y)\cdot 2x - (1)(x^2+y^2)}{(x+y)^2}$$

$$= \frac{2x^2 + 2xy - x^2 - y^2}{(x+y)^2} = \frac{x^2 + 2xy - y^2}{(x+y)^2}$$

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Differentiating eqn (1) partially w.r.to y we get

$$\frac{\partial z}{\partial y} = \frac{\partial}{\partial y}\left[\frac{(x^2+y^2)}{(x+y)}\right]$$

$$= \frac{(x+y)(2y) - (1)(x^2+y^2)}{(x+y)^2}$$

$$= \frac{-x^2+2xy+y^2}{(x+y)^2}$$

$$\left(\frac{\partial z}{\partial x} - \frac{\partial z}{\partial y}\right) = \left[\frac{x^2+2xy-y^2}{(x+y)^2}\right] - \left[\frac{-x^2+2xy+y^2}{(x+y)^2}\right]$$

$$= \frac{2x^2+2y^2}{(x+y)^2} = 2\frac{(x^2-y^2)}{(x+y)^2}$$

$$= 2\frac{(x-y)}{(x+y)}$$

$$\left(\frac{\partial z}{\partial x} - \frac{\partial z}{\partial y}\right)^2 = 4\frac{(x-y)^2}{(x+y)^2} \quad \text{——— (2)}$$

$$4\left(1 - \frac{\partial z}{\partial x} - \frac{\partial z}{\partial y}\right) = 4\left(1 - \frac{x^2+2xy-y^2}{(x+y)^2} - \frac{-x^2+2xy+y^2}{(x+y)^2}\right)$$

$$= 4\frac{(x^2-2xy+y^2)}{(x+y)^2}$$

$$= 4\frac{(x-y)^2}{(x+y)^2} \quad \text{——— (3)}$$

from eqn (2) and (3), we have

$$\left(\frac{\partial z}{\partial x} - \frac{\partial z}{\partial y}\right)^2 = 4\left(1 - \frac{\partial z}{\partial x} - \frac{\partial z}{\partial y}\right).$$

Hyp.

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Ex: Find the value of n so that the equation $V = r^n(3\cos^2\theta - 1)$ satisfy the relation.

$$\frac{\partial}{\partial r}\left(r^2 \frac{\partial V}{\partial r}\right) + \frac{1}{\sin\theta} \frac{\partial}{\partial \theta}\left(\sin\theta \cdot \frac{\partial V}{\partial \theta}\right) = 0$$

Soln. Given

$$V = r^n(3\cos^2\theta - 1) \quad \text{——— (1)}$$

Differentiating (1) partially w.r.to r, we get

$$\frac{\partial V}{\partial r} = n \cdot r^{n-1}(3\cos^2\theta - 1)$$

$$\Rightarrow r^2\left(\frac{\partial V}{\partial r}\right) = n\, r^{n-1} r^2 (3\cos^2\theta - 1)$$

$$= n\, r^{n+1} (3\cos^2\theta - 1) \quad \text{——— (2)}$$

Differentiating again partially w.r.to r, we get

$$\frac{\partial}{\partial r}\left(r^2 \frac{\partial V}{\partial r}\right) = n\cdot(n+1)\, r^n (3\cos^2\theta - 1)$$

$$= n(n+1) \cdot V$$

$$= (n^2+n)V \quad \text{——— (3)} \quad (\because \text{by (1)})$$

Now. Differentiating (1) partially w.r.to θ, we get

$$\frac{\partial V}{\partial \theta} = r^n [6\cos\theta(-\sin\theta)]$$

$$= -6\, r^n \cos\theta \sin\theta$$

$$\therefore \sin\theta\left(\frac{\partial V}{\partial \theta}\right) = -6\, r^n \cos\theta \sin^2\theta \quad \text{——— (4)}$$

Differentiating again w.r.to θ. we get

$$\frac{\partial}{\partial \theta}\left(\sin\theta \cdot \frac{\partial V}{\partial \theta}\right) = -6 r^n \frac{\partial}{\partial \theta}(\cos\theta \cdot \sin^2\theta)$$

$$= -6 r^n (-\sin^3\theta + 2\sin\theta \cdot \cos^2\theta)$$

$$= -6 r^n \sin\theta (-\sin^2\theta + 2\cos^2\theta)$$

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$$= -6r^n \sin\theta [-(1-\cos^2\theta) + 2\cos^2\theta]$$

$$= -6r^n \sin\theta [3\cos^2\theta - 1]$$

$$\frac{\partial}{\partial\theta}\left(\sin\theta \frac{\partial v}{\partial\theta}\right) = -6\sin\theta \cdot v$$

$$\Rightarrow \frac{1}{\sin\theta} \cdot \frac{\partial}{\partial\theta}\left(\sin\theta \frac{\partial v}{\partial\theta}\right) = -6v \quad \text{——(5)}$$

Now, since v satisfy the given relation therefor by adding (3) and (5) equating to zero we get

$$\frac{\partial}{\partial r}\left(r^2 \frac{\partial v}{\partial r}\right) + \frac{1}{\sin\theta} \frac{\partial}{\partial\theta}\left(\sin\theta \frac{\partial v}{\partial\theta}\right) = 0$$

$$\Rightarrow (n^2+n)v - 6v = 0$$

$$\Rightarrow n^2+n-6 = 0$$

$$\Rightarrow (n+3)(n-2) = 0$$

$$\Rightarrow \boxed{n = -3, 2}$$

$\therefore$ for the value $n = -3$, or 2 we satisfy the given relation

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Ex: If $x^x y^y z^z = c$ show that at $x=y=z$

$$\frac{\partial^2 z}{\partial x \partial y} = -(x \log e x)^{-1} \quad \text{or}$$

$$= -\frac{1}{x}(1+\log x)^{-1}$$

Soln: Given that

$$x^x y^y z^z = c$$

Taking log on both side we get.

$$x \log x + y \log y + z \log z = \log c. \quad \text{---(1)}$$

Differentiating (1) partially w.r.to x. assuming z as a function of x and y. we get

$$\left(x\frac{1}{x} + \log x\right) + 0 + \left(z\frac{1}{z}\frac{\partial z}{\partial x} + \log z \cdot \frac{\partial z}{\partial x}\right) = 0$$

$$\Rightarrow (1+\log x) + \left(\frac{\partial z}{\partial x} + \log z \frac{\partial z}{\partial x}\right) = 0$$

$$\Rightarrow \frac{\partial z}{\partial x}(1+\log z) = -(1+\log x)$$

$$\Rightarrow \frac{\partial z}{\partial x} = -\frac{(1+\log x)}{(1+\log z)} \quad \text{---(2)}$$

Similarly differentiating (1) partialy w.r.to. y .we get.

$$\frac{\partial z}{\partial y} = -\frac{(1+\log y)}{(1+\log z)} \quad \text{---(3)}$$

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Now diff. eqn ② partially w.r.to y. we get

$$\frac{\partial^2 z}{\partial y \partial x} = \frac{\partial}{\partial y}\left(\frac{\partial z}{\partial x}\right) = \frac{\partial}{\partial y}\left[\frac{-(1+\log x)}{(1+\log z)}\right]$$

$$= -(1+\log x)\cdot\left[\frac{-1}{(1+\log z)^2}\right]\cdot\frac{1}{z}\cdot\frac{\partial z}{\partial y}$$

$$= \frac{(1+\log x)}{(1+\log z)^2}\cdot\frac{1}{z}\left[\frac{-(1+\log y)}{(1+\log z)}\right]$$

$$\frac{\partial^2 z}{\partial y \partial x} = -\frac{(1+\log x)(1+\log y)}{z\,(1+\log z)^3}$$

Now at $x = y = z$

$$\frac{\partial^2 z}{\partial y \partial x} = -\frac{(1+\log y)(1+\log x)}{x(1+\log x)^3}$$

$$= -\frac{1}{x}\,\frac{1}{(1+\log x)}$$

$$= -\frac{1}{x}(1+\log x)^{-1}$$

$$= -\frac{1}{x}\left[\log e + \log x\right]^{-1} \qquad \because \log_e e = 1$$

$$= -\frac{1}{x}\left[\log(ex)\right]^{-1}$$

$$\therefore \frac{\partial^2 z}{\partial y \partial x} = -\frac{1}{x}\left[\log(ex)\right]^{-1}$$

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Ex: ① prove that If $f(x,y) = \frac{1}{\sqrt{y}} e^{-\frac{(x-a)^2}{4y}}$

$f_{xy} = f_{yx}$. $\left[f_{xy} = \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial x}\right) \right.$
$\left. f_{yx} = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial y}\right) \right.$

② If $u = x^y$. show that $\frac{\partial^3 u}{\partial x^2 \partial y} = \frac{\partial^2 u}{\partial x \partial y \cdot \partial x}$

③ If $v = At^{-1/2} e^{-x^2/4a^2t}$. prove that

$$\frac{\partial v}{\partial t} = a^2 \frac{\partial^2 v}{\partial x^2}$$

④ If $z = e^{(ax+by)} f(ax+by)$

prove that $b\left(\frac{\partial z}{\partial x}\right) + a\left(\frac{\partial z}{\partial y}\right) = 2abz$.

⑤ If $z = x^y + y^x$, prove that $\frac{\partial^2 z}{\partial x \partial y} = \frac{\partial^2 z}{\partial y \partial x}$

Hint.

$z = x^y + y^x$
$= e^{\log x^y} + e^{\log y^x}$
$= e^{y \log x} + e^{x \log y}$.

⑥ If $z = x^2 \tan^{-1}\left(\frac{y}{x}\right) - y^2 \tan^{-1}\left(\frac{x}{y}\right)$

prove that

$$\frac{\partial^2 z}{\partial x \partial y} = \frac{\partial^2 z}{\partial y \partial x} = \frac{x^2 - y^2}{x^2 + y^2}$$

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Homogeneous function :⇒

A function $f(x,y)$ is said to be homogeneous function in which the power of each term is same are

A function $f(x,y)$ is a homogeneous function of order n. If the degree of each of its terms in x and y is equal to n. therfore

$$f(x,y) = a_0x^n + a_1x^{n-1}y + a_2x^{n-2}y^2 + \cdots + a_ny^n \quad \text{---(1)}$$

is homogeneous function of order n.

The polynomial function (1) which also can be written as

$$f(x,y) = x^n\left[a_0 + a_1\left(\frac{y}{x}\right) + a_2\left(\frac{y}{x}\right)^2 + \cdots + a_n\left(\frac{y}{x}\right)^n\right] = x^n\,\phi\left(\frac{y}{x}\right)$$

The function $x^3\left[1+\left(\frac{y}{x}\right)+3\left(\frac{y}{x}\right)^2+5\left(\frac{y^3}{x^3}\right)\right]$ is homogeneous function of order 3.

$$\frac{\sqrt{x}+\sqrt{y}}{x^2+y^2} = \frac{\sqrt{x}\left(1+\sqrt{\frac{y}{x}}\right)}{x^2\left(1+\left(\frac{y}{x}\right)^2\right)} = \frac{x^{1/2}\left[1+\left(\frac{y}{x}\right)^{1/2}\right]}{x^2\left[1+\left(\frac{y}{x}\right)^2\right]} = x^{-3/2}\,\phi\left(\frac{y}{x}\right)$$

is homogeneous function of order $-3/2$

) $\sin^{-1}\left(\frac{\sqrt{x}+\sqrt{y}}{x^2+y^2}\right)$ is not a homogeneous function as it can not be written in the form $x^n\phi\left(\frac{y}{x}\right)$. so its degree ~~is~~ may be pronounced.

i.e. if $z = \sin^{-1}\left(\frac{\sqrt{x}+\sqrt{y}}{x^2+y^2}\right)$

$\Rightarrow \sin z = \frac{\sqrt{x}+\sqrt{y}}{x^2+y^2} = x^{-3/2}\,\phi\left(\frac{y}{x}\right)$ $\therefore \sin z$ is homogeneous fun of degree $-3/2$.

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EULER'S Thm on Homogeneous function

Statement: If z is a homogeneous function of x, y of order n then

$$x\frac{\partial z}{\partial x} + y\frac{\partial z}{\partial y} = n z$$

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* I^st deduction of Euler's Thm

If z is a homogeneous function of x, y of degree n and $z = f(u)$ then

$$x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = n\frac{f(u)}{f'(u)}$$

* II^nd deduction of Euler's Thm :⇒

If z is homogeneous function of x, y of degree n and $z = f(u)$ then

$$x^2\frac{\partial^2 u}{\partial x^2} + 2xy\frac{\partial^2 u}{\partial x \partial y} + y^2\frac{\partial^2 u}{\partial y^2} = g(u)\,[g'(u) - 1]$$

where, $\boxed{g(u) = n\frac{f(u)}{f'(u)}}$

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Ex: If z be a homogeneous function of degree n show that

(i) $x\frac{\partial^2 z}{\partial x^2} + y\frac{\partial^2 z}{\partial x \partial y} = (n-1)\frac{\partial z}{\partial x}$

(ii) $x\frac{\partial^2 z}{\partial x \partial y} + y\frac{\partial^2 z}{\partial y^2} = (n-1)\frac{\partial z}{\partial y}$

(iii) $x^2\frac{\partial^2 z}{\partial x^2} + 2xy\frac{\partial^2 z}{\partial x \partial y} + y^2\frac{\partial^2 z}{\partial y^2} = n(n-1)z$

Soln: (i) let z is homogeneous function of degree n.

$\therefore$ By Euler's thm. we have,

$$x\frac{\partial z}{\partial x} + y\frac{\partial z}{\partial y} = nz \quad \text{——(1)}$$

Diff. eqn (1) partialy w.r.to. x. we get

$$\left[x\frac{\partial^2 z}{\partial x^2} + (1)\frac{\partial z}{\partial x}\right] + y\frac{\partial^2 z}{\partial x \partial y} = n\cdot\frac{\partial z}{\partial x}$$

$$\Rightarrow x\frac{\partial^2 z}{\partial x^2} + y\frac{\partial^2 z}{\partial x \partial y} = n\frac{\partial z}{\partial x} - \frac{\partial z}{\partial x}$$

$$\Rightarrow \boxed{x\frac{\partial^2 z}{\partial x^2} + y\frac{\partial^2 z}{\partial x \partial y} = (n-1)\frac{\partial z}{\partial x}} \quad \text{——(2)}$$

H.P.

(ii) Diff eqn (1) partialy w.r.to. y. we get

$$x\frac{\partial^2 z}{\partial y \partial x} + \left[y\frac{\partial^2 z}{\partial y^2} + \frac{\partial z}{\partial y}\right] = n\frac{\partial z}{\partial y}$$

$$\Rightarrow \boxed{x\frac{\partial^2 z}{\partial x \partial y} + y\frac{\partial^2 z}{\partial y^2} = (n-1)\frac{\partial z}{\partial y}} \quad \text{——(3)}$$

H.P

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Now multiplying eqn (2) by x and (3) by y and then adding we get

$$\left(x^2\frac{\partial^2 z}{\partial x^2} + xy\frac{\partial^2 z}{\partial x \partial y}\right) + \left(xy\frac{\partial^2 z}{\partial x \partial y} + y^2\frac{\partial^2 z}{\partial y^2}\right)$$

$$= n(n-1)\cdot x\cdot\frac{\partial z}{\partial x} + (n-1)\,y\frac{\partial z}{\partial y}$$

$$= (n-1)\left[x\frac{\partial z}{\partial x} + y\frac{\partial z}{\partial y}\right]$$

$$= n\cdot(n-1)\cdot z \qquad (\because \text{by (1)})$$

$$\therefore \boxed{x^2\frac{\partial^2 z}{\partial x^2} + 2xy\frac{\partial^2 z}{\partial x \partial y} + y^2\frac{\partial^2 z}{\partial y^2} = n(n-1)z}$$

H.P.

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Ex: If $z = x^n f\left(\frac{y}{x}\right) + y^{-n}\phi\left(\frac{x}{y}\right)$ (8)

Then prove that

$$x^2\frac{\partial^2 z}{\partial x^2} + 2xy\frac{\partial^2 z}{\partial x \partial y} + y^2\frac{\partial^2 z}{\partial y^2} + x\frac{\partial z}{\partial x} + y\frac{\partial z}{\partial y} = n^2 z$$

Soln: ⇒ Let

$$z = x^n f\left(\frac{y}{x}\right) + y^{-n}\phi\left(\frac{x}{y}\right)$$

$$\Rightarrow z = u + v \quad \text{——— (1)}$$

where

$$u = x^n f\left(\frac{y}{x}\right)$$

$$v = y^{-n}\phi\left(\frac{x}{y}\right)$$

where, u, and v are the ~~function of~~ homogeneous function. in x and y of degree n and −n respectively.

∴ By Eulers thm we have

$$x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = n\cdot u \quad \text{——— (2)}$$

and $$x^2\frac{\partial^2 u}{\partial x^2} + 2xy\frac{\partial^2 u}{\partial x \partial y} + y^2\frac{\partial^2 u}{\partial y^2} = n(n-1)u \quad \text{——— (3)}$$

Similarly $$x\frac{\partial v}{\partial x} + y\frac{\partial v}{\partial y} = n\cdot v \quad \text{——— (4)}$$

& $$x^2\frac{\partial^2 v}{\partial x^2} + 2xy\frac{\partial^2 v}{\partial x \partial y} + y^2\frac{\partial^2 v}{\partial y^2} = n(n-1)v \quad \text{——— (5)}$$

adding eqn (2) and (4) we get

$$\left(x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y}\right) + \left(x\frac{\partial v}{\partial x} + y\frac{\partial v}{\partial y}\right) = nu + nv$$

$$\Rightarrow x\left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial x}\right) + y\left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial y}\right) = n(u+v)$$

$$\Rightarrow x\frac{\partial}{\partial x}(u+v) + y\frac{\partial}{\partial y}(u+v) = n(u+v)$$

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$$\therefore \quad x\frac{\partial z}{\partial x} + y\frac{\partial z}{\partial y} = nz \qquad (\because \text{by } (1))$$

⑥

again adding ③ and ④ we get

$$\left(x^2\frac{\partial^2 u}{\partial x^2} + 2xy\frac{\partial^2 u}{\partial x\partial y} + y^2\frac{\partial^2 u}{\partial y^2}\right) + \left(x^2\frac{\partial^2 v}{\partial x^2} + 2xy\frac{\partial^2 v}{\partial x\partial y} + y^2\frac{\partial^2 v}{\partial y^2}\right)$$

$$= n(n-1)u + n(n-1)v$$

$$\Rightarrow \quad x^2\frac{\partial^2}{\partial x^2}(u+v) + 2xy\frac{\partial^2 (u+v)}{\partial x\partial y} + y^2\frac{\partial^2}{\partial y^2}(u+v) = n(n-1)(u+v)$$

$$\Rightarrow \quad x^2\frac{\partial^2 z}{\partial x^2} + 2xy\frac{\partial^2 z}{\partial x\partial y} + y^2\frac{\partial^2 z}{\partial y^2} = n(n-1)z \qquad \text{by (2)}$$

⑦

Now adding eqn ⑥ and ⑦ we get

$$x^2\frac{\partial^2 z}{\partial x^2} + 2xy\frac{\partial^2 z}{\partial x\partial y} + y^2\frac{\partial^2 z}{\partial y^2} + x\frac{\partial z}{\partial x} + y\frac{\partial z}{\partial y} = nz + n(n-1)z$$

$$= nz + n^2z - nz$$

$$= n^2z$$

$$\therefore \boxed{x^2\frac{\partial^2 z}{\partial x^2} + 2xy\frac{\partial^2 z}{\partial x\partial y} + y^2\frac{\partial^2 z}{\partial y^2} + x\frac{\partial z}{\partial x} + y\frac{\partial z}{\partial y} = n^2z}$$

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Ex: If $u = f(v)$, v being homogeneous of n^{th} degree in x and y then prove that

$$x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = nv \cdot f'(v)$$ and hence if $u = \log v$ then prove that $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = n$

Soln :⇒ Let v is homogeneous function of x & y of degree n.

∴ By Euler's thm., we have

$$x\frac{\partial v}{\partial x} + y\frac{\partial v}{\partial y} = nv \quad \text{——(1)}$$

Now let $u = f(v)$ ——(2)

Differentiating eqn (2) w.r.to x partially, we get.

$$\frac{\partial u}{\partial x} = f'(v) \cdot \frac{\partial v}{\partial x}$$

$$\Rightarrow x\frac{\partial u}{\partial x} = f'(v)\, x\frac{\partial v}{\partial x} \quad \text{——(3)}$$

IIⁿᵈ Differentiating eqn (2) partialy w.r to y we get

$$\frac{\partial u}{\partial y} = f'(v) \cdot \frac{\partial v}{\partial y}$$

$$\Rightarrow y\frac{\partial u}{\partial y} = f'(v) \cdot y\frac{\partial v}{\partial y} \quad \text{——(4)}$$

Now adding eqn (3) and (4) we get

$$x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = f'(v)\left(x\frac{\partial v}{\partial x} + y\frac{\partial v}{\partial y}\right)$$

$$= f'(v) \cdot n \cdot v \quad (\because \text{by } (1))$$

$$\therefore \boxed{x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = n \cdot v \cdot f'(v)} \quad \text{——(5)}$$

Hence proved

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Now let $u = \log v$

i.e $f(v) = \log v$

$\therefore f'(v) = \frac{1}{v}$

Substituting this value in eqn (5), we get

$$x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = n \cdot v \cdot \left(\frac{1}{v}\right)$$

$$\Rightarrow \boxed{x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = n.}$$

Ex:⇒ If $u = \tan^{-1}\left(\frac{x^3+y^3}{x-y}\right)$, prove that $\frac{(tx)^3+(ty)^3}{(tx-ty)}$

(i) $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = \sin 2u$

(ii) $x^2\frac{\partial^2 u}{\partial x^2} + 2xy\frac{\partial^2 u}{\partial x \partial y} + y^2\frac{\partial^2 u}{\partial y^2} = 2\cos 3u \cdot \sin$

Soln. let $u = \tan^{-1}\left(\frac{x^3+y^3}{x-y}\right)$

is not a homogeneous function.

But we can see that

$$\tan u = \frac{x^3+y^3}{x-y} = \frac{x^3\left[1+\left(\frac{y}{x}\right)^3\right]}{x\left[1-\left(\frac{y}{x}\right)\right]} = x^2\,\phi(\frac{}{}$$

i.e. $z = \tan(u) = f(u)$ is a homogeneous function of degree $\underline{\underline{n=2}}$

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∴ By Euler's 1st deduction we have

$$x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = n\cdot\frac{f(u)}{f'(u)}$$

Here $f(u) = \tan u$

$\therefore f'(u) = \sec^2 u$ and $n = 2$

$$\therefore x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = 2\frac{\tan u}{\sec^2 u}$$

$$= 2\frac{\sin u}{\cos u}\cdot\cos^2 u$$

$$= 2\sin u\cos u$$

$$\therefore \boxed{x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = \sin 2u}$$

Now. By Euler's 2nd deduction we have

$$x^2\frac{\partial^2 u}{\partial x^2} + 2xy\frac{\partial^2 u}{\partial x\partial y} + y^2\frac{\partial^2 u}{\partial y^2} = g(u)[g'(u) - 1]$$

where. $g(u) = n\frac{f(u)}{f'(u)} = \sin 2u$

$\Rightarrow g'(u) = 2\cos 2u$

$$\therefore x^2\frac{\partial^2 u}{\partial x^2} + 2xy\frac{\partial^2 u}{\partial x\partial y} + y^2\frac{\partial^2 u}{\partial y^2} = 2\sin 2u\cdot\cos[2\cos 2u - 1]$$

$$= 2\sin 2u\cos 2u - \sin 2u$$

$$= \sin 4u - \sin 2u$$

$$= \sin(3u+u) - \sin(3u-u)$$

$$\boxed{x^2\frac{\partial^2 u}{\partial x^2} + 2xy\frac{\partial^2 u}{\partial x\partial y} + y^2\frac{\partial^2 u}{\partial y^2} = 2\cos 3u\cdot\sin u}$$

$\therefore \cos A\sin B = \frac{1}{2}[\sin(A+B) - \sin(A-B)]$

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Ex: If $u = \sin^{-1}\left[\frac{x+y}{\sqrt{x}+\sqrt{y}}\right]$ prove that

$$x^2\frac{\partial^2 u}{\partial x^2} + 2xy\frac{\partial^2 u}{\partial x\partial y} + y^2\frac{\partial^2 u}{\partial y^2} = -\frac{\sin u\cdot\cos 2u}{4\cos^3 u}.$$

Soln: ⇒ let $u = \sin^{-1}\left(\frac{x+y}{\sqrt{x}+\sqrt{y}}\right)$ —①

is not a homogeneous, but we can see that

$$z = \sin u = \left(\frac{x+y}{\sqrt{x}+\sqrt{y}}\right) = \frac{x\left[1+\left(\frac{y}{x}\right)\right]}{\sqrt{x}\left(1+\sqrt{\frac{y}{x}}\right)}$$

$$= x^{1/2}\,\phi\left(\frac{y}{x}\right)$$

∴ is homogeneous function of dgree $n = 1/2$

let $z = \sin u = f(u)$.

Then by Euler 2nd deduction we have

$$x^2\frac{\partial^2 u}{\partial x^2} + 2xy\frac{\partial^2 u}{\partial x\partial y} + y^2\frac{\partial^2 u}{\partial y^2} = g(u)\left[g'(u)-1\right]$$

where, $g(u) = n\frac{f(u)}{f'(u)}$

But $f(u) = \sin u$

$\therefore f'(u) = \cos u$ and $n = 1/2$

$$\therefore g(u) = \frac{1}{2}\frac{\sin u}{\cos u} = \frac{1}{2}\tan u$$

$$\Rightarrow g'(u) = \frac{1}{2}\sec^2 u$$

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$$\therefore x^2\frac{\partial^2 u}{\partial x^2} + 2xy\frac{\partial^2 u}{\partial x\partial y} + y^2\frac{\partial^2 u}{\partial y^2} = \frac{1}{2}\tan u\left[\frac{1}{2}\sec^2 u - 1\right] \quad (11)$$

$$= \frac{1}{2}\tan u\left[\frac{\sec^2 u - 2}{2}\right]$$

$$= \frac{1}{4}\left[\tan u \cdot \sec^2 u - 2\tan u\right]$$

$$= \frac{1}{4}\left[\frac{\sin u}{\cos u}\cdot\frac{1}{\cos^2 u} - 2\frac{\sin u}{\cos u}\right]$$

$$= \frac{1}{4}\left[\frac{\sin u}{\cos^3 u} - \frac{2\sin u}{\cos u}\right]$$

$$= \frac{1}{4}\left[\frac{\sin u - 2\sin u\cos^2 u}{\cos^3 u}\right]$$

$$= \frac{1}{4}\frac{\sin u}{\cos 3u}\left[1 - 2\cos^2 u\right]$$

$$= \frac{1}{4}\cdot\frac{\sin u}{\cos^3 u}\cdot\left[-(2\cos^2 u - 1)\right]$$

$$= -\frac{1}{4}\frac{\sin u\cdot\sin 2u}{\cos^3 u}$$

$$\therefore \boxed{x^2\frac{\partial^2 u}{\partial x^2} + 2xy\frac{\partial^2 u}{\partial x\partial y} + y^2\frac{\partial^2 u}{\partial y^2} = -\frac{1}{4}\frac{\sin u\cdot\sin 2u}{\cos^3 u}}$$

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Ex: If $u = \log\left(\frac{x^2+y^2}{\sqrt{x}+\sqrt{y}}\right)$. then find the value of

(i) $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y}$ (ii) $x^2\frac{\partial^2 u}{\partial x^2} + 2xy\frac{\partial^2 u}{\partial x \partial y} + y^2$

Soln: ⇒ Let

$$u = \log\left(\frac{x^2+y^2}{\sqrt{x}+\sqrt{y}}\right)$$

is not a homogeneous function. But

$$z = \log e^u = \left(\frac{x^2+y^2}{\sqrt{x}+\sqrt{y}}\right) = \frac{x^2\left[1+\left(\frac{y}{x}\right)^2\right]}{\sqrt{x}\left[1+\sqrt{\frac{y}{x}}\right]}$$

$$= x^{3/2}\,\phi\left(\frac{y}{x}\right)$$

i's homogeneous function of degree $n = 3/2$

where $z = \log e^u = f(u)$

∴ By Euler's 1st deduction we have

$$x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = n\frac{f(u)}{f'(u)}$$

Here. $f(u) = \log e^u$

$\Rightarrow f'(u) = e^u$ and $n = 3/2$

$$\therefore x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = \frac{3}{2}\cdot\frac{\log u}{1/u}\,\frac{e^u}{e^u}$$

$$\Rightarrow \boxed{\left(x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y}\right) = 3/2}$$

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Now by 4th deduction of Euler's thm. we have (12)

$$x^2\frac{\partial^2 u}{\partial x^2} + 2xy\frac{\partial^2 u}{\partial x\partial y} + y^2\frac{\partial^2 u}{\partial y^2} = g(u)[g'(u)-1]$$

where

$$g(u) = n\frac{f(u)}{f'(u)} = 3/2$$

$$\therefore g'(u) = 0$$

$$\therefore \quad x^2\frac{\partial^2 u}{\partial x^2} + 2xy\frac{\partial^2 u}{\partial y\partial x} + y^2\frac{\partial^2 u}{\partial y^2} = 3/2[0-1]$$

$$= -3/2$$

Ex :⇒ If $u = x^2\phi\left(\frac{y}{x}\right) + y\,\psi\left(\frac{x}{y}\right)$

Find the value of $x^2\frac{\partial^2 u}{\partial x^2} + 2xy\frac{\partial u}{\partial x\partial y} + y^2\frac{\partial^2 u}{\partial y^2}$

Soln:

let $u = x^2\phi\left(\frac{y}{x}\right) + y\,\psi\left(\frac{x}{y}\right)$

$$\therefore u = u_1 + u_2 \qquad \text{---(1)}$$

where

$$u_1 = x^2\phi\left(\frac{y}{x}\right)$$

& $u_2 = y\,\phi\left(\frac{x}{y}\right)$

Here u_1 and u_2 both are the homogeneous function of degree $n=2$ and $n=1$ respectively

$\therefore$ By Euler's Thm. we have

$$x^2\frac{\partial^2 u_1}{\partial x^2} + 2xy\frac{\partial u_1}{\partial x\partial y} + y^2\frac{\partial^2 u_1}{\partial y^2} = n(n-1)u_1$$

$$= 2(2-1)u_1$$

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$$\therefore x^2\frac{\partial^2 u_1}{\partial x^2} + 2xy\frac{\partial^2 u_1}{\partial x \partial y} + y^2\frac{\partial^2 u_1}{\partial y^2} = 2u_1$$

$$= 2x^2\phi\left(\frac{y}{x}\right)$$

and $\because$ for $u_2 = y\,\psi\left(\frac{y}{x}\right)$.

$$x^2\frac{\partial^2 u_2}{\partial x^2} + 2xy\frac{\partial^2 u_2}{\partial x \partial y} + y^2\frac{\partial^2 u_2}{\partial y^2} = 1(1-1)u_2 = 0$$

Now adding eqn. (2) and (3) we get

$$x^2\left(\frac{\partial^2 u_1}{\partial x^2} + \frac{\partial^2 u_2}{\partial x^2}\right) + 2xy\left(\frac{\partial^2 u_1}{\partial x \partial y} + \frac{\partial^2 u_2}{\partial x \partial y}\right) + y^2\left(\frac{\partial^2 u_1}{\partial y^2} + \frac{\partial^2 u_2}{\partial y^2}\right)$$

$$= 2x^2\phi\left(\frac{y}{x}\right) +$$

$$\Rightarrow x^2\frac{\partial^2}{\partial x^2}(u_1+u_2) + 2xy\frac{\partial^2(u_1+u_2)}{\partial x \partial y} + y^2\frac{\partial^2(u_1+u_2)}{\partial y^2}$$

$$= 2x^2\phi\left(\frac{y}{x}\right)$$

$$\Rightarrow \boxed{x^2\frac{\partial^2 u}{\partial x^2} + 2xy\frac{\partial^2 u}{\partial x \partial y} + y^2\frac{\partial^2 u}{\partial y^2} = 2x^2\phi\left(\frac{y}{x}\right)}$$

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Ex: If $u = \frac{(x^2+y^2)^n}{2n(2n-1)} + x\phi\left(\frac{y}{x}\right) + \psi\left(\frac{y}{x}\right)$. (18)

Then show that

$$x^2\frac{\partial^2 u}{\partial x^2} + 2xy\frac{\partial^2 u}{\partial x \partial y} + y^2\frac{\partial^2 u}{\partial y^2} = (x^2+y^2)^n.$$

Soln: let

$$u = \frac{(x^2+y^2)^n}{2n(2n-1)} + x\phi\left(\frac{y}{x}\right) + \psi\left(\frac{y}{x}\right)$$

$$\therefore u = u_1 + u_2 + u_3 \quad \text{——— (1)}$$

where

$$u_1 = \frac{(x^2+y^2)^n}{2n(2n-1)} = \frac{x^{2n}\left[1+\left(\frac{y}{x}\right)^2\right]^n}{2n(2n-1)}$$

$$u_2 = x^1\phi\left(\frac{y}{x}\right)$$

& $$u_3 = x^0\psi\left(\frac{y}{x}\right)$$

are homogeneous function of degree $2n$, 1 and 0 respectively.

$\therefore$ By Euler's thm. we have

For u_1

$$x^2\frac{\partial^2 u_1}{\partial x^2} + 2xy\frac{\partial^2 u_1}{\partial x \partial y} + y^2\frac{\partial^2 u_1}{\partial y^2} = n(n-1)u_1$$

(n is degree of u_1)

$$= 2n(2n-1)\cdot\frac{(x^{2n}+y^2)^n}{2n(2n-1)}$$

$$= (x^2+y^2)^n \quad \text{——— (2)}$$

Similarly For u_2

$$x^2\frac{\partial^2 u_2}{\partial x^2} + 2xy\frac{\partial^2 u_2}{\partial x \partial y} + y^2\frac{\partial^2 u_2}{\partial y^2} = n(n-1)u_2$$

$$= 1(1-1)\left[x\phi\left(\frac{y}{x}\right)\right] = 0 \quad \text{——— (3)}$$

For u_3

& $$x^2\frac{\partial^2 u_3}{\partial x^2} + 2xy\frac{\partial^2 u_3}{\partial x \partial y} + y^2\frac{\partial^2 u_3}{\partial y^2} = 0(0-1)\cdot u_3 = 0 \quad \text{——— (4)}$$

adding eqn (2) (3) and (4) we get.

$$x^2\frac{\partial^2 u}{\partial x^2} + 2xy\frac{\partial^2 u}{\partial x \partial y} + y^2\frac{\partial^2 u}{\partial y^2} = (x^2+y^2)^n.$$

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Ex: If $u = \frac{x^3+y^3}{y\sqrt{x}} + \frac{1}{x^7}\sin^{-1}\left(\frac{x^2+y^2}{x^2+2xy}\right)$

find value of

$$x^2\frac{\partial^2 u}{\partial x^2} + 2xy\frac{\partial^2 u}{\partial x \partial y} + y^2\frac{\partial^2 u}{\partial y^2} + x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y}$$

at $x=1,\ y=2.$

Sol:⇒ let

$$u = \frac{x^3+y^3}{y\sqrt{x}} + \frac{1}{x^7}\sin^{-1}\left(\frac{x^2+y^2}{x^2+2xy}\right)$$

$$u = u_1 + u_2 \qquad \text{---(1)}$$

where.

$$u_1 = \frac{x^3+y^3}{y\sqrt{x}}$$

$$= \frac{x^3\left[1+(y/x)^3\right]}{x^{1/2}\cdot y}$$

$$= x^{5/2}\frac{\left[1+(y/x)^3\right]}{y} = x^{3/2}\cdot x\frac{\left[1+(\frac{y}{x})^3\right]}{y}$$

$$= x^{3/2}\cdot\frac{\left[1+(\frac{y}{x})^3\right]}{(y/x)} = x^{3/2}\phi\left(\frac{y}{x}\right)$$

and

$$u_2 = \frac{1}{x^7}\sin^{-1}\left(\frac{x^2+y^2}{x^2+2xy}\right)$$

$$= x^{-7}\sin^{-1}\left[\frac{x^2\left(1+(y/x)^2\right]}{x^2\left[1+2\frac{y}{x}\right]}\right]$$

$$= x^{-7}\sin^{-1}\left[\frac{1+(y/x)^2}{(1+2y/x)}\right]$$

$$= x^{-7}\cdot\phi\left(\frac{y}{x}\right)$$

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This shows that, u_1 and u_2 homogeneous function (14) of degree $3/2$ and -7 respectively

$\therefore$ By Eulers Thm. we have

$$x^2\frac{\partial^2 u_1}{\partial x^2} + 2xy\frac{\partial^2 u_1}{\partial x\partial y} + y^2\frac{\partial^2 u_1}{\partial y^2} = n(n-1)u_1$$

$$= 3/2(3/2-1)\cdot u_1$$

$$= \frac{3}{4}u_1$$

and $$x\frac{\partial u_1}{\partial x} + y\frac{\partial u_2}{\partial x_2} = n\cdot u_1 \quad = 3/2\, u_1$$

$$\therefore x^2\frac{\partial^2 u_1}{\partial x^2} + 2xy\frac{\partial^2 u_1}{\partial x\partial y} + y^2\frac{\partial^2 u_1}{\partial y^2} + x\frac{\partial u_1}{\partial x} + y\frac{\partial u_1}{\partial y} = \frac{3}{4}u_1 + \frac{3}{2}u_1$$

$$= \frac{9}{4}u_1 \qquad \text{——— (2)}$$

Similarly

$$x^2\frac{\partial^2 u_2}{\partial x^2} + 2xy\frac{\partial^2 u_2}{\partial x\partial y} + y^2\frac{\partial^2 u_2}{\partial y^2} + x\frac{\partial u_2}{\partial x} + y\frac{\partial u_2}{\partial y} = -7(-7-1)u_2 + (-7)u_2$$

$$= 49\,u_2 \qquad \text{——— (3)}$$

adding eqn (2) and (3) and using (1) we get

$$x^2\frac{\partial^2 u}{\partial x^2} + 2xy\frac{\partial^2 u}{\partial x\partial y} + y^2\frac{\partial^2 u}{\partial y^2} + x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = \frac{9}{4}u_1 + 49u_2 \qquad \text{——— (4)}$$

Now at $x = 1$ and $y = 2$

$$u_1 = \frac{x^3+y^3}{y\sqrt{x}} = \frac{1+8}{2\sqrt{1}} = \frac{9}{2}$$

$$u_2 = \frac{1}{x^7}\sin^{-1}\left(\frac{x^2+y^2}{x^2+2xy}\right) = \frac{1}{(1)^7}\cdot\sin^{-1}\left[\frac{1+4}{1+2(1)(2)}\right]$$

$$= \sin^{-1}(1) = \pi/2$$

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$$\therefore x^2\frac{\partial^2 u}{\partial x^2} + 2xy\frac{\partial^2 u}{\partial x\partial y} + y^2\frac{\partial^2 u}{\partial y^2} = \frac{9}{4}\cdot\left(\frac{9}{2}\right) + 49\cdot(\pi/2)$$

$$= \frac{81}{8} + \frac{49\pi}{2}$$

H.A

(1) If $u = \tan^{-1}(y^2/x)$ then find the value of

$$x^2\frac{\partial^2 u}{\partial x^2} + 2xy\frac{\partial^2 u}{\partial x\partial y} + y^2\frac{\partial^2 u}{\partial y^2}$$

$\frac{1}{2}\sin 2u(\cos 2u - 1)$

(2) If $u = \sin^{-1}\left[\frac{x^{1/3} + y^{1/3}}{x^{1/2} + y^{1/2}}\right]^{1/2}$.

Find the value of: $x^2\frac{\partial^2 u}{\partial x^2} + 2xy\frac{\partial^2 u}{\partial x\partial y} + y^2\frac{\partial^2 u}{\partial y^2}$

$= \frac{1}{12}\tan u\left[-\frac{13}{12} + \frac{\tan^2 u}{12}\right]$

(3) If $u = \tan^{-1}\left[\frac{x^{1/2} + y^{1/2}}{x^{1/3} - y^{1/3}}\right]$ find the value of

(i) $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = \frac{1}{12}\sin 2u$

(ii) $x^2\frac{\partial^2 u}{\partial x^2} + 2xy\frac{\partial^2 u}{\partial x\partial y} + y^2\frac{\partial^2 u}{\partial y^2} = \frac{1}{144}\sin 4u - \frac{1}{12}\sin$

Proved

(4) If $u = \log(x^4 + x^2y^2 + y^4)$

Show that $x^2\frac{\partial^2 u}{\partial x^2} + 2xy\frac{\partial^2 u}{\partial x\partial y} + y^2\frac{\partial^2 u}{\partial y^2} + 4 = 0$

(5) if $u = \log\left(\frac{x^3+y^3}{x+y}\right)$ find the value of

$x^2\frac{\partial^2 u}{\partial x^2} + 2xy\frac{\partial^2 u}{\partial x\partial y} + y^2\frac{\partial^2 u}{\partial y^2} = 0$

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Total Differentiation

Let $z = f(x, y)$ ——— (1)

be a function any two variable x and y.

let δx, δy be the small increments in x and y respectively then corresponding increment in z is δz

$$\therefore z + \delta z = f(x + \delta x, y + \delta y) \quad \text{——— (2)}$$

Substracting eqn (1) from (2), we get

$$\delta z = f(x+\delta x, y+\delta y) - f(x, y)$$

$$= f(x+\delta x, y+\delta y) - f(x+\delta x, y) + f(x+\delta x, y) - f(x, y)$$

$$= \frac{f(x+\delta x, y+\delta y) - f(x+\delta x, y)}{\delta y} \cdot \delta y + \frac{f(x+\delta x, y) - f(x, y)}{\delta x} \cdot \delta x$$

$$\therefore \lim_{\substack{\delta x \to 0 \\ \delta y \to 0}} \delta z = \lim_{\substack{\delta y \to 0 \\ \delta x \to 0}} \frac{f(x+\delta x, y+\delta y) - f(x+\delta x, y)}{\delta y} \cdot \lim_{\substack{\delta y \to 0 \\ \delta x \to 0}} \delta y + \lim_{\substack{\delta x \to 0 \\ \delta y \to 0}} \frac{f(x+\delta x, y) - f(x, y)}{\delta x} \cdot \lim_{\substack{\delta x \to 0 \\ \delta y \to 0}} \delta x$$

$$\Rightarrow dz = \frac{\partial f}{\partial y} \cdot dy + \frac{\partial f}{\partial x} \cdot dx$$

i.e $$\boxed{dz = \frac{\partial z}{\partial x} dx + \frac{\partial z}{\partial y} dy}$$

is called the total differential of z

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Chain rule of partial differentiation :-

If $z = f(x,y)$ and $x = \phi(t)$, $y = \psi(t)$

where t is a parameter then we can express z as a function of t alone.

Hence we can regard z as a function of the single variable t and can find the ordinary differential coefficient $\frac{dz}{dt}$ which is called the total derivative of z and is given by

$$\boxed{\frac{dz}{dt} = \frac{\partial z}{\partial x}\cdot\frac{dx}{dt} + \frac{\partial z}{\partial y}\cdot\frac{dy}{dt}}$$

In case of three independent variables we can write

$$\boxed{\frac{du}{dt} = \frac{\partial u}{\partial x}\frac{dx}{dt} + \frac{\partial u}{\partial y}\frac{dy}{dt} + \frac{\partial u}{\partial z}\frac{dz}{dt}}$$

if $u = u(x,y,z)$

This is called the chain rule of partial differentiation

If $z = f(x,y)$

and $x = \phi(u,v)$, and $y = \psi(u,v)$

Then

$$\boxed{\frac{\partial z}{\partial u} = \frac{\partial z}{\partial x}\cdot\frac{\partial x}{\partial u} + \frac{\partial z}{\partial y}\cdot\frac{\partial y}{\partial u}}$$

&

$$\boxed{\frac{\partial z}{\partial v} = \frac{\partial z}{\partial x}\cdot\frac{\partial x}{\partial v} + \frac{\partial z}{\partial y}\cdot\frac{\partial y}{\partial v}}$$

Important

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Ex: If $z = f(x,y)$, where, $x = e^u \cos v$ & $y = e^u \sin v$ (15)

Show that

$$y\frac{\partial z}{\partial u} + x\frac{\partial z}{\partial v} = e^{2u}\frac{\partial z}{\partial y}$$

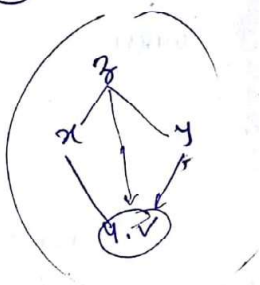
Soln: let

$$z = f(x,y)$$

and

$$x = e^u \cos v = \phi(u,v) \quad \text{——— (1)}$$

$$y = e^u \sin v = \psi(u,v) \quad \text{——— (2)}$$



Thus z is function of (u,v).

$$\therefore \frac{\partial z}{\partial u} = \frac{\partial z}{\partial x}\cdot\frac{\partial x}{\partial u} + \frac{\partial z}{\partial y}\cdot\frac{\partial y}{\partial u} \quad \text{——— (3)}$$

$$\& \quad \frac{\partial z}{\partial v} = \frac{\partial z}{\partial x}\cdot\frac{\partial x}{\partial v} + \frac{\partial z}{\partial y}\cdot\frac{\partial y}{\partial v} \quad \text{——— (4)}$$

Now Diff. eqn (1) and (2) w.r.to u and v, we get

$$\frac{\partial x}{\partial u} = e^u \cos v, \quad = x \qquad \frac{\partial x}{\partial v} = e^u(-\sin v) = -y$$

$$\frac{\partial y}{\partial u} = e^u \sin v = y, \quad \& \quad \frac{\partial y}{\partial v} = e^u \cos v = x$$

$\therefore$ eqn (3) $\Rightarrow$ $\frac{\partial z}{\partial u} = x\frac{\partial z}{\partial x} + y\frac{\partial z}{\partial y}$

$$\Rightarrow y\frac{\partial z}{\partial u} = xy\frac{\partial z}{\partial x} + y^2\frac{\partial z}{\partial y} \quad \text{——— (5)}$$

Now eqn (4) $\Rightarrow$ $\frac{\partial z}{\partial v} = -y\frac{\partial z}{\partial x} + x\frac{\partial z}{\partial y}$

$$\Rightarrow x\frac{\partial z}{\partial v} = -xy\frac{\partial z}{\partial x} + x^2\frac{\partial z}{\partial y} \quad \text{——— (6)}$$

adding eqn (5) and (6) we get

$$y\frac{\partial z}{\partial u} + x\frac{\partial z}{\partial v} = y^2\frac{\partial z}{\partial y} + x^2\frac{\partial z}{\partial y}$$

$$= (e^u \sin v)^2\frac{\partial z}{\partial y} + (e^u \cos v)^2\frac{\partial z}{\partial y}$$

$$= e^{2u}(\sin^2 v + \cos^2 v)\frac{\partial z}{\partial y} = e^{2u}\frac{\partial z}{\partial y} \quad \#$$

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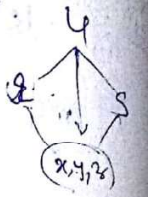
Ex: If $u = u\left(\frac{y-x}{xy}, \frac{z-x}{xz}\right)$.

Show that

$$x^2 \frac{\partial u}{\partial x} + y^2 \frac{\partial u}{\partial y} + z^2 \frac{\partial u}{\partial z} = 0$$

Soln: $\Rightarrow$ Let $u = u\left(\frac{y-x}{xy}, \frac{z-x}{xz}\right)$

$\Rightarrow u = u(\xi, \zeta)$



where $\xi = \frac{y-x}{xy} = \phi(x,y,z)$ & $\zeta = \frac{z-x}{xz} = \psi(x,y,z)$

$\xi = \frac{1}{x} - \frac{1}{y}$

$\zeta = \frac{1}{x} - \frac{1}{z}$

This shows that u is composite function of x, y, z.

$$\therefore \frac{\partial u}{\partial x} = \frac{\partial u}{\partial \xi}\cdot\frac{\partial \xi}{\partial x} + \frac{\partial u}{\partial \zeta}\cdot\frac{\partial \zeta}{\partial x} \quad \text{——(1)}$$

$$\frac{\partial u}{\partial y} = \frac{\partial u}{\partial \xi}\cdot\frac{\partial \xi}{\partial y} + \frac{\partial u}{\partial \zeta}\cdot\frac{\partial \zeta}{\partial y} \quad \text{——(2)}$$

$$\& \quad \frac{\partial u}{\partial z} = \frac{\partial u}{\partial \xi}\cdot\frac{\partial \xi}{\partial z} + \frac{\partial u}{\partial \zeta}\cdot\frac{\partial \zeta}{\partial z} \quad \text{——(3)}$$

Now, we have.

$$\frac{\partial \xi}{\partial x} = \frac{-1}{x^2}, \quad \frac{\partial \xi}{\partial y} = \frac{1}{y^2}, \quad \frac{\partial \xi}{\partial z} = 0$$

$$\frac{\partial \zeta}{\partial x} = \frac{-1}{x^2}, \quad \frac{\partial \zeta}{\partial y} = 0, \quad \frac{\partial \zeta}{\partial z} = \frac{1}{z^2}$$

Substituting these value in eqn (1) (2) and (3), we get

(1) $\Rightarrow \frac{\partial u}{\partial x} = \frac{-1}{x^2}\frac{\partial u}{\partial \xi} + \frac{1}{x^2}\frac{\partial u}{\partial \zeta} \Rightarrow x^2\frac{\partial u}{\partial x} = -\frac{\partial u}{\partial \xi} - \frac{\partial u}{\partial \zeta}$ —(A)

(2) $\Rightarrow \frac{\partial u}{\partial y} = \frac{1}{y^2}\frac{\partial u}{\partial \xi} + 0 \Rightarrow y^2\frac{\partial u}{\partial y} = \frac{\partial u}{\partial \xi}$ —(B)

(3) $\Rightarrow \frac{\partial u}{\partial z} = 0 + \frac{1}{z^2}\cdot\frac{\partial u}{\partial \zeta} \Rightarrow z^2\frac{\partial u}{\partial z} = \frac{\partial u}{\partial \zeta}$ —(C)

Adding (A) (B) and (C), we get

$$\boxed{x^2\frac{\partial u}{\partial x} + y^2\frac{\partial u}{\partial y} + z^2\frac{\partial u}{\partial z} = 0}$$

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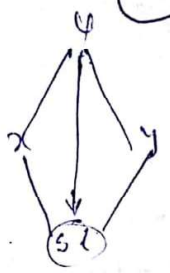
Q. If $u = f(x, y)$

and $x = s\cos\alpha - t\sin\alpha$

$y = s\sin\alpha + t\cos\alpha$



prove that $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = \frac{\partial^2 u}{\partial s^2} + \frac{\partial^2 u}{\partial t^2}$

Soln: Let $u = f(x, y)$

and $x = s\cos\alpha - t\sin\alpha = \phi(s, t)$

$y = s\sin\alpha + t\cos\alpha = \psi(s, t)$

$\Rightarrow$ u is composite function of (s, t).

$$\therefore \frac{\partial u}{\partial t} = \frac{\partial u}{\partial x}\cdot\frac{\partial x}{\partial t} + \frac{\partial u}{\partial y}\cdot\frac{\partial y}{\partial t} \quad \text{——— (1)}$$

$$\text{and } \frac{\partial u}{\partial s} = \frac{\partial u}{\partial x}\cdot\frac{\partial x}{\partial s} + \frac{\partial u}{\partial y}\cdot\frac{\partial y}{\partial s} \quad \text{——— (2)}$$

Now, we have

$$\frac{\partial x}{\partial t} = -\sin\alpha, \quad \frac{\partial x}{\partial s} = \cos\alpha$$

$$\frac{\partial y}{\partial t} = \cos\alpha, \quad \frac{\partial y}{\partial s} = \sin\alpha$$

$$\therefore (1) \Rightarrow \frac{\partial u}{\partial t} = -\sin\alpha\frac{\partial u}{\partial x} + \cos\alpha\frac{\partial u}{\partial y} \quad \text{——— (3)}$$

$$\Rightarrow \frac{\partial}{\partial t}u = \left(-\sin\alpha\cdot\frac{\partial}{\partial x} + \cos\alpha\frac{\partial}{\partial y}\right)u$$

$$\Rightarrow \frac{\partial}{\partial t} = \left(-\sin\alpha\frac{\partial}{\partial x} + \cos\alpha\frac{\partial}{\partial y}\right) \quad \text{——— (4)}$$

$$\therefore \frac{\partial^2 u}{\partial t^2} = \frac{\partial}{\partial t}\cdot\left(\frac{\partial u}{\partial t}\right)$$

$$= \left(-\sin\alpha\frac{\partial}{\partial x} + \cos\alpha\frac{\partial}{\partial y}\right)\left(-\sin\alpha\frac{\partial u}{\partial x} + \cos\alpha\frac{\partial u}{\partial y}\right)$$

$$= \sin^2\alpha\frac{\partial}{\partial x}\left(\frac{\partial u}{\partial x}\right) - \sin\alpha\cos\alpha\cdot\frac{\partial}{\partial x}\frac{\partial u}{\partial y} - \sin\alpha\cos\alpha\frac{\partial}{\partial y}\frac{\partial u}{\partial x} + \cos^2\frac{\partial}{\partial y}\cdot\frac{\partial u}{\partial y}$$

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$$= \sin^2\frac{\partial^2 u}{\partial x^2} + \sin\alpha\cos\alpha\frac{\partial^2 u}{\partial x\partial y} + \sin\alpha\cos\alpha\frac{\partial^2 u}{\partial y\partial x} + \cos^2\frac{\partial^2 u}{\partial y^2}$$

$$\frac{\partial^2 u}{\partial t^2} = \sin^2\frac{\partial^2 u}{\partial x^2} + 2\sin\alpha\cos\alpha\frac{\partial^2 u}{\partial x\partial y} + \cos^2\frac{\partial^2 u}{\partial y^2}$$

Now,

$$\frac{\partial u}{\partial s^2} = \frac{\partial}{\partial s} \quad \text{from eqn (2)}.$$

$$\frac{\partial u}{\partial s} = \cos\alpha\frac{\partial u}{\partial x} + \sin\alpha\frac{\partial u}{\partial y} \quad \text{---(5)}$$

$$\Rightarrow \frac{\partial}{\partial s} = \left(\cos\alpha\frac{\partial}{\partial x} + \sin\alpha\frac{\partial}{\partial y}\right) \quad \text{---(6)}$$

$$\therefore \frac{\partial^2 u}{\partial s^2} = \frac{\partial}{\partial s}\left(\frac{\partial u}{\partial s}\right)$$

$$= \left(\cos\alpha\frac{\partial}{\partial x} + \sin\alpha\frac{\partial}{\partial y}\right)\cdot\left(\cos\alpha\frac{\partial u}{\partial x} + \sin\alpha\frac{\partial u}{\partial y}\right)$$

$$= \cos^2\alpha\frac{\partial^2 u}{\partial x^2} + 2\sin\alpha\cos\alpha\frac{\partial^2 u}{\partial x\partial y} + \sin^2\alpha\frac{\partial^2 u}{\partial y^2} \quad \text{---(B)}$$

Now adding eqn (A) and (B) we get

$$\frac{\partial^2 u}{\partial t^2} + \frac{\partial^2 u}{\partial s^2} = (\sin^2\alpha + \cos^2\alpha)\frac{\partial^2 u}{\partial x^2} + 0 + (\sin^2\alpha + \cos^2\alpha)\frac{\partial^2 u}{\partial y^2}$$

$$= \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}$$

$$\therefore \boxed{\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = \frac{\partial^2 u}{\partial t^2} + \frac{\partial^2 u}{\partial s^2}}$$

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Ex: If $x+y = 2e^{\theta}\cos\phi$ & $x-y = 2i\,e^{\theta}\sin\phi$

Show that

$$\frac{\partial^2 V}{\partial \theta^2} + \frac{\partial^2 V}{\partial \phi^2} = 4xy\,\frac{\partial^2 V}{\partial x\,\partial y}$$

Sol: let $V = f(x,y)$ —(1)

and it is given that

$$x+y = 2e^{\theta}\cos\phi \quad \text{—(2)}$$

$$x-y = 2i\,e^{\theta}\sin\phi \quad \text{—(3)}$$

adding eqn (2) and (3) we get

$$2x = 2e^{\theta}\cos\phi + 2i\,e^{\theta}\sin\phi$$

$$= 2e^{\theta}(\cos\phi + i\sin\phi) = 2e^{\theta}e^{i\phi}$$

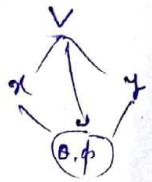
$$\Rightarrow x = e^{\theta+i\phi} \quad \text{—(4)}$$

Substracting eqn (2) and (3) we get

$$2y = 2e^{\theta}(\cos\phi - i\sin\phi)$$

$$= 2e^{\theta}e^{-i\phi}$$

$$\Rightarrow y = e^{\theta-i\phi} \quad \text{—(5)}$$



It is clear that V is function of x and y and x and y both are the function of θ and φ

i.e V is composite function of θ and φ

$$\therefore \frac{\partial V}{\partial \theta} = \frac{\partial V}{\partial x}\cdot\frac{\partial x}{\partial \theta} + \frac{\partial V}{\partial y}\cdot\frac{\partial y}{\partial \theta} \quad \text{—(6)}$$

&

$$\frac{\partial V}{\partial \phi} = \frac{\partial V}{\partial x}\cdot\frac{\partial x}{\partial \phi} + \frac{\partial V}{\partial y}\cdot\frac{\partial y}{\partial \phi} \quad \text{—(7)}$$

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Differentiating eqn ④ and ⑤ partially w.r.t. θ and ϕ we get

from ④ $$\frac{\partial x}{\partial \theta} = e^{\theta + i\phi} = x$$

$$\frac{\partial x}{\partial \phi} = e^{\theta + i\phi}(i) = ix$$

from ⑤ $$\frac{\partial y}{\partial \theta} = e^{\theta - i\phi} = y$$

$$\frac{\partial y}{\partial \phi} = (-i)e^{\theta - i\phi} = -iy$$

Now from eqn ⑥ we get

$$\frac{\partial v}{\partial \theta} = \frac{\partial v}{\partial x}\cdot\frac{\partial x}{\partial \theta} + \frac{\partial v}{\partial y}\cdot\frac{\partial y}{\partial \theta}$$

$$\therefore \frac{\partial v}{\partial \theta} = x\frac{\partial v}{\partial x} + y\frac{\partial v}{\partial y}$$

$$\Rightarrow \frac{\partial}{\partial \theta} = \left(x\frac{\partial}{\partial x} + y\frac{\partial}{\partial y}\right)$$

$$\therefore \frac{\partial^2 v}{\partial \theta^2} = \frac{\partial}{\partial \theta}\left(\frac{\partial v}{\partial \theta}\right)$$

$$= \left(x\frac{\partial}{\partial x} + y\frac{\partial}{\partial y}\right)\left[x\frac{\partial v}{\partial x} + y\frac{\partial v}{\partial y}\right]$$

$$= x\frac{\partial}{\partial x}\left(x\frac{\partial v}{\partial x} + y\frac{\partial v}{\partial y}\right) + y\frac{\partial}{\partial y}\left(x\frac{\partial v}{\partial x} + y\frac{\partial v}{\partial y}\right)$$

$$= x\left[x\frac{\partial^2 v}{\partial x^2} + (1)\frac{\partial v}{\partial x} + y\frac{\partial^2 v}{\partial x \partial y}\right]$$

$$+ y\left[x\frac{\partial^2 v}{\partial x \partial y} + y\frac{\partial^2 v}{\partial y^2} + (1)\frac{\partial v}{\partial y}\right]$$

$$\frac{\partial^2 v}{\partial \theta^2} = x^2\frac{\partial^2 v}{\partial x^2} + 2xy\frac{\partial^2 v}{\partial x \partial y} + y^2\frac{\partial^2 v}{\partial y^2} + x\frac{\partial v}{\partial x} + y\frac{\partial v}{\partial y}$$

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Now. from eqn (7) we have

$$\frac{\partial \phi}{\partial r} = \frac{\partial x}{\partial r}\cdot\frac{\partial \phi}{\partial x} + \frac{\partial y}{\partial r}\cdot\frac{\partial \phi}{\partial y}$$

$$= \quad = i x \frac{\partial r}{\partial x} - i y \frac{\partial r}{\partial y}$$

$$\frac{\partial \phi}{\partial r} = i\left(x\frac{\partial r}{\partial x} - y\frac{\partial r}{\partial y}\right)$$

$$\Rightarrow \frac{\partial \phi}{\partial r} = i\left(\frac{x}{r}x - \frac{y}{r}y\right)$$

$$\therefore \frac{\partial^2 \phi}{\partial r^2} = \frac{\partial}{\partial r}\left(\frac{\partial \phi}{\partial r}\right)$$

$$= i\left(\frac{x}{r}x - \frac{y}{r}y\right) \times i\left(\frac{x}{r}\frac{\partial r}{\partial x} - \frac{y}{r}\frac{\partial r}{\partial y}\right) =$$

$$= (-1)\left\{x\frac{\partial}{\partial r}\left[x\frac{\partial r}{\partial x} - y\frac{\partial r}{\partial y}\right] - y\frac{\partial}{\partial r}\left[x\frac{\partial r}{\partial x} - y\frac{\partial r}{\partial y}\right]\right\} \quad (i^2 = -1)$$

$$= -\left\{x\left[x\frac{\partial^2 r}{\partial x^2} + \frac{\partial r}{\partial x} - y\frac{\partial^2 r}{\partial x \partial y}\right] - y\left[x\frac{\partial^2 r}{\partial x \partial y} - y\frac{\partial^2 r}{\partial y^2}\right]\right\}$$

$$= -\left[x^2\frac{\partial^2 r}{\partial x^2} + x\frac{\partial r}{\partial x} - xy\frac{\partial^2 r}{\partial x \partial y} - xy\frac{\partial^2 r}{\partial x \partial y} + y^2\frac{\partial^2 r}{\partial y^2} + y\frac{\partial r}{\partial y}\right]$$

$$\frac{\partial^2 \phi}{\partial r^2} = -x^2\frac{\partial^2 r}{\partial x^2} - x\frac{\partial r}{\partial x} + 2xy\frac{\partial^2 r}{\partial x \partial y} - y^2\frac{\partial^2 r}{\partial y^2} - y\frac{\partial r}{\partial y} \quad \longrightarrow (9)$$

adding eqn (8) and (9) we get

$$\boxed{\frac{\partial^2 v}{\partial \theta^2} + \frac{\partial^2 v}{\partial \phi^2} = 4xy\frac{\partial^2 v}{\partial x \partial y}}$$

H.W.

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Ex If $\phi = \phi(x,y,z)$ &

If $x = \sqrt{vw}$, $y = \sqrt{wu}$, $z = \sqrt{uv}$

and ϕ is a function of x, y and z

then prove that

$$x\frac{\partial\phi}{\partial x} + y\frac{\partial\phi}{\partial y} + z\frac{\partial\phi}{\partial z} = u\frac{\partial\phi}{\partial u} + v\frac{\partial\phi}{\partial v} + w$$

Soln :⇒ Let $x = \sqrt{vw} = f(u,v,w)$ —①

$y = \sqrt{wu} = g(u,v,w)$ —②

$z = \sqrt{uv} = h(u,v,w)$ —③

$\Rightarrow \frac{\partial x}{\partial u} = 0$, $\frac{\partial x}{\partial v} = \frac{\sqrt{w}}{2\sqrt{v}} = \frac{1}{2}\sqrt{\frac{w}{v}}$

$\frac{\partial x}{\partial w} = \frac{1}{2}\sqrt{\frac{v}{w}}$

$\frac{\partial y}{\partial u} = \frac{1}{2}\sqrt{\frac{w}{u}}$, $\frac{\partial y}{\partial v} = 0$, $\frac{\partial y}{\partial w} = \frac{1}{2}\sqrt{\frac{u}{w}}$

$\frac{\partial z}{\partial u} = \frac{1}{2}\sqrt{\frac{v}{u}}$, $\frac{\partial z}{\partial v} = \frac{1}{2}\sqrt{\frac{u}{v}}$, $\frac{\partial z}{\partial w} = 0$

It is clear that ϕ is a function of x, y, z and x, y, z are function of u, v and w.

i.e. ϕ is composite function of u, v, w.

∴ we have

$$\frac{\partial\phi}{\partial u} = \frac{\partial\phi}{\partial x}\cdot\frac{\partial x}{\partial u} + \frac{\partial\phi}{\partial y}\frac{\partial y}{\partial u} + \frac{\partial\phi}{\partial z}\cdot\frac{\partial z}{\partial u}$$

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$$\therefore \frac{\partial \phi}{\partial u} = 0 \cdot \frac{\partial \phi}{\partial x} + \frac{1}{2}\sqrt{\frac{w}{u}} \cdot \frac{\partial \phi}{\partial y} + \frac{1}{2}\sqrt{\frac{v}{u}} \frac{\partial \phi}{\partial z}$$

$$= \frac{1}{2}\left(\sqrt{\frac{w}{u}} \cdot \frac{\partial \phi}{\partial y} + \sqrt{\frac{v}{u}} \frac{\partial \phi}{\partial z}\right)$$

$$\therefore u \frac{\partial \phi}{\partial u} = \frac{1}{2} u \frac{\sqrt{w}}{\sqrt{u}} \frac{\partial \phi}{\partial y} + \frac{1}{2} u \frac{\sqrt{v}}{\sqrt{u}} \frac{\partial \phi}{\partial z}$$

$$= \frac{1}{2}\sqrt{uw} \frac{\partial \phi}{\partial y} + \frac{1}{2}\sqrt{uv} \frac{\partial \phi}{\partial z}$$

$$u \frac{\partial \phi}{\partial u} = \frac{1}{2} y \frac{\partial \phi}{\partial y} + \frac{z}{2} \frac{\partial \phi}{\partial z} \quad \text{——— (4)}$$

(ii)

$$\frac{\partial \phi}{\partial v} = \frac{\partial \phi}{\partial x} \cdot \frac{\partial x}{\partial v} + \frac{\partial \phi}{\partial y} \cdot \frac{\partial y}{\partial v} + \frac{\partial \phi}{\partial z} \cdot \frac{\partial z}{\partial v}$$

$$= \frac{1}{2}\sqrt{\frac{w}{v}} \cdot \frac{\partial \phi}{\partial x} + 0 \cdot \frac{\partial \phi}{\partial y} + \frac{1}{2}\sqrt{\frac{u}{v}} \frac{\partial \phi}{\partial z}$$

$$\Rightarrow \frac{\partial \phi}{\partial v} = \frac{1}{2}\frac{\sqrt{w}}{\sqrt{v}} \frac{\partial \phi}{\partial x} + \frac{1}{2}\frac{\sqrt{u}}{\sqrt{v}} \frac{\partial \phi}{\partial z}$$

$$\therefore v \frac{\partial \phi}{\partial v} = \frac{1}{2}\sqrt{vw} \frac{\partial \phi}{\partial x} + \frac{1}{2}\sqrt{vu} \frac{\partial \phi}{\partial z}$$

$$v \frac{\partial \phi}{\partial v} = \frac{x}{2} \frac{\partial \phi}{\partial x} + \frac{z}{2} \frac{\partial \phi}{\partial z} \quad \text{——— (5)}$$

(iii)

$$\frac{\partial \phi}{\partial w} = \frac{\partial \phi}{\partial x} \frac{\partial x}{\partial w} + \frac{\partial \phi}{\partial y} \cdot \frac{\partial y}{\partial w} + \frac{\partial \phi}{\partial z} \cdot \frac{\partial z}{\partial w}$$

$$= \frac{1}{2}\sqrt{\frac{v}{w}} \frac{\partial \phi}{\partial x} + \frac{1}{2}\sqrt{\frac{u}{w}} \frac{\partial \phi}{\partial y} + 0 \frac{\partial \phi}{\partial z}$$

$$\frac{\partial \phi}{\partial w} = \frac{1}{2}\frac{\sqrt{v}}{\sqrt{w}} \frac{\partial \phi}{\partial x} + \frac{1}{2}\frac{\sqrt{u}}{\sqrt{w}} \frac{\partial \phi}{\partial y}$$

$$\therefore w \frac{\partial \phi}{\partial w} = \frac{1}{2}\sqrt{wv} \frac{\partial \phi}{\partial x} + \frac{1}{2}\sqrt{wu} \frac{\partial \phi}{\partial y} = \frac{x}{2} \frac{\partial \phi}{\partial x} + \frac{y}{2} \frac{\partial \phi}{\partial y} \quad \text{——— (6)}$$

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adding eqn. (4), (5) and (6) we get.

$$u\frac{\partial\phi}{\partial u} + v\frac{\partial\phi}{\partial v} + w\frac{\partial\phi}{\partial w} = \left(\frac{y}{2}\frac{\partial\phi}{\partial y} + \frac{z}{2}\frac{\partial\phi}{\partial z}\right) + \left(\frac{x}{2}\frac{\partial\phi}{\partial x} + \frac{z}{2}\frac{\partial\phi}{\partial z}\right) + \left(\frac{x}{2}\frac{\partial\phi}{\partial x} + \frac{y}{2}\frac{\partial\phi}{\partial y}\right)$$

$$= x\frac{\partial\phi}{\partial x} + y\frac{\partial\phi}{\partial y} + z\frac{\partial\phi}{\partial z}$$

$$\therefore \boxed{u\frac{\partial\phi}{\partial u} + v\frac{\partial\phi}{\partial v} + w\frac{\partial\phi}{\partial w} = x\frac{\partial\phi}{\partial x} + y\frac{\partial\phi}{\partial y} + z\frac{\partial\phi}{\partial z}}$$

Hence proved

Ex: If $z = f(u,v)$ and u and v are two other functions such that $u = lx + my$ and $v = ly - mx$ show that

$$\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} = (l^2 + m^2)\left(\frac{\partial^2 z}{\partial u^2} + \frac{\partial^2 z}{\partial v^2}\right)$$

Soln: $\Rightarrow$ let $z = f(u,v)$.

and $u = lx + my$

$v = ly - mx$

both are function of x and y.

i.e. z is composite function function of x and y

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$$\therefore \frac{\partial z}{\partial x} = \frac{\partial z}{\partial u}\cdot\frac{\partial u}{\partial x} + \frac{\partial z}{\partial v}\cdot\frac{\partial v}{\partial x} \quad \text{---(1)}$$

and $$\frac{\partial z}{\partial y} = \frac{\partial z}{\partial u}\cdot\frac{\partial u}{\partial y} + \frac{\partial z}{\partial v}\cdot\frac{\partial v}{\partial y} \quad \text{---(2)}$$

Now Here.

$$\frac{\partial u}{\partial x} = l, \quad \frac{\partial u}{\partial y} = m$$

$$\frac{\partial v}{\partial x} = -m, \quad \frac{\partial v}{\partial y} = l$$

$$\therefore \frac{\partial z}{\partial x} = l\frac{\partial z}{\partial u} + (-m)\frac{\partial z}{\partial v}$$

$$\Rightarrow \frac{\partial}{\partial x} = \left(l\frac{\partial}{\partial u} - m\frac{\partial}{\partial v}\right)$$

$$\therefore \frac{\partial^2 z}{\partial x^2} = \frac{\partial}{\partial x}\left(\frac{\partial z}{\partial x}\right)$$

$$= \left(l\frac{\partial}{\partial u} - m\frac{\partial}{\partial v}\right)\cdot\left[l\frac{\partial z}{\partial u} - m\frac{\partial z}{\partial v}\right]$$

$$= l^2\frac{\partial}{\partial u}\left(\frac{\partial z}{\partial u}\right) - lm\frac{\partial}{\partial u}\left(\frac{\partial z}{\partial v}\right) - ml\frac{\partial}{\partial v}\left(\frac{\partial z}{\partial u}\right) + m^2\frac{\partial}{\partial v}\left(\frac{\partial z}{\partial v}\right)$$

$$\frac{\partial^2 z}{\partial x^2} = l^2\frac{\partial^2 z}{\partial u^2} - 2ml\frac{\partial^2 z}{\partial u\partial v} + m^2\frac{\partial^2 z}{\partial v^2} \quad \text{---(3)}$$

Now

$$\frac{\partial z}{\partial y} = m\frac{\partial z}{\partial u} + l\frac{\partial z}{\partial v}$$

$$\therefore \frac{\partial}{\partial y} = \left(m\frac{\partial}{\partial u} + l\frac{\partial}{\partial v}\right)$$

$$\therefore \frac{\partial^2 z}{\partial y^2} = \frac{\partial}{\partial y}\left(\frac{\partial z}{\partial y}\right) = \left(m\frac{\partial}{\partial u} + l\frac{\partial}{\partial v}\right)\left(m\frac{\partial z}{\partial u} + l\frac{\partial z}{\partial v}\right)$$

$$= m^2\frac{\partial^2 z}{\partial u^2} + 2lm\frac{\partial^2 z}{\partial u\partial v} + l^2\frac{\partial^2 z}{\partial v^2} \quad \text{---(4)}$$

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adding eqn ③ and ④ we get

$$\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} = l^2\left(\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2}\right)$$

$$= \left(l^2 \frac{\partial^2 z}{\partial u^2} + m^2 \frac{\partial^2 z}{\partial y^2}\right)$$

$$+ \left(m^2 \frac{\partial^2 z}{\partial u^2} + l^2 \frac{\partial^2 z}{\partial v^2}\right)$$

$$= l^2\left(\frac{\partial^2 z}{\partial u^2} + \frac{\partial^2 z}{\partial v^2}\right) + m^2\left(\frac{\partial^2 z}{\partial v^2} + \right.$$

$$= (l^2+m^2)\left(\frac{\partial^2 z}{\partial u^2} + \frac{\partial^2 z}{\partial v^2}\right)$$

$$\therefore \boxed{\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} = (l^2+m^2)\left(\frac{\partial^2 z}{\partial u^2} + \frac{\partial^2 z}{\partial v^2}\right)}$$

H.P.

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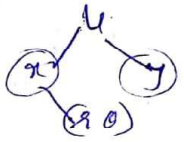
$u = f(x, y)$

If $x = r\cos\theta$, $y = r\sin\theta$, prove that-

$$x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} + y^2\frac{\partial^2 u}{\partial x^2} - 2xy\frac{\partial^2 u}{\partial x \partial y} + x^2\frac{\partial^2 u}{\partial y^2} = r\frac{\partial u}{\partial r} + \frac{\partial^2 u}{\partial \theta^2}$$



Sol^n: let

$x = r\cos\theta$

$y = r\sin\theta$

both are the function of r and θ.

∴ Differentiating x and y partialy w.r.to r and θ we get

$$\frac{\partial x}{\partial r} = \cos\theta, \quad \frac{\partial x}{\partial \theta} = -r\sin\theta$$

$$\frac{\partial y}{\partial r} = \sin\theta, \quad \frac{\partial y}{\partial \theta} = r\cos\theta$$

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Now, let u is a function of x and y but x and y are function of r, θ

i.e. u is composite function of (r, θ)

$$\therefore \frac{\partial u}{\partial r} = \frac{\partial u}{\partial x}\cdot\frac{\partial x}{\partial r} + \frac{\partial u}{\partial y}\frac{\partial y}{\partial r}$$

$$\Rightarrow \frac{\partial u}{\partial r} = \cos\theta \frac{\partial u}{\partial x} + \sin\theta \frac{\partial u}{\partial y} \quad \text{(I)}$$

$$\Rightarrow \frac{\partial}{\partial r} = \cos\theta \frac{\partial}{\partial x} + \sin\theta \frac{\partial}{\partial y}$$

$$\therefore \frac{\partial^2 u}{\partial r^2} \Rightarrow r\frac{\partial u}{\partial r} = r\cos\theta \frac{\partial u}{\partial x} + r\sin\theta \frac{\partial u}{\partial y} \quad \text{(II)}$$

and $$\frac{\partial u}{\partial \theta} = \frac{\partial u}{\partial x}\cdot\frac{\partial x}{\partial \theta} + \frac{\partial u}{\partial y}\cdot\frac{\partial y}{\partial \theta}$$

$$\Rightarrow \frac{\partial u}{\partial \theta} = -r\sin\theta \frac{\partial u}{\partial x} + r\cos\theta \frac{\partial u}{\partial y}$$

$$\Rightarrow \frac{\partial}{\partial \theta} = \left(-r\sin\theta \frac{\partial}{\partial x} + r\cos\theta \frac{\partial}{\partial y}\right)$$

$$\frac{\partial^2 u}{\partial \theta^2} = \frac{\partial}{\partial \theta}\left(\frac{\partial u}{\partial \theta}\right)$$

$$= \left(-r\sin\theta \frac{\partial}{\partial x} + r\cos\theta \frac{\partial}{\partial y}\right)\left(-r\sin\theta \frac{\partial u}{\partial x} + r\cos\theta \frac{\partial u}{\partial y}\right)$$

$$= r^2\sin^2\theta \frac{\partial^2 u}{\partial x^2} - r^2\sin\theta\cos\theta \frac{\partial^2 u}{\partial x \partial y} - r^2\cos\theta\sin\theta \frac{\partial^2}{\partial}$$

$$+ r^2\cos^2\theta \frac{\partial^2 u}{\partial y^2}$$

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$$\Rightarrow \frac{\partial^2 u}{\partial \theta^2} = r^2 \sin^2\theta \frac{\partial^2 u}{\partial x^2} - 2r^2 \sin\theta\cos\theta \frac{\partial^2 u}{\partial x \partial y} + r^2 \cos^2\theta \frac{\partial^2 u}{\partial y^2} \quad \text{(2)}$$

Now adding eqn (1) and (2) we get

$$r\frac{\partial u}{\partial r} + \frac{\partial^2 u}{\partial \theta^2} = r\cos\theta \frac{\partial u}{\partial x} + r\sin\theta \frac{\partial u}{\partial y} + r^2\sin^2\theta \frac{\partial^2 u}{\partial x^2} + 2r^2 \sin\theta\cos\theta \frac{\partial^2 u}{\partial x \partial y} + r^2\cos^2\theta \frac{\partial^2 u}{\partial y^2}$$

$$= x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} + y^2 \frac{\partial^2 u}{\partial x^2} - 2xy \frac{\partial^2 u}{\partial x \partial y} + x^2 \frac{\partial^2 u}{\partial y^2}$$

$\because \left(x = r\cos\theta,\ y = r\sin\theta \right)$

$$\therefore \boxed{x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} + y^2 \frac{\partial^2 u}{\partial x^2} - 2xy \frac{\partial^2 u}{\partial x \partial y} + x^2 \frac{\partial^2 u}{\partial y^2} = r\frac{\partial u}{\partial r} + \frac{\partial^2 u}{\partial \theta^2}}$$

Hence proved

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Ex: If $x = r\cos\theta$, $y = r\sin\theta$ then prove that

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = \frac{\partial^2 u}{\partial r^2} + \frac{1}{r}\frac{\partial u}{\partial r} + \frac{1}{r^2}\frac{\partial^2 u}{\partial \theta^2}$$

Soln: Let us consider

$u = u(r, \theta)$ be the function of (r, θ).

and $x = r\cos\theta$, $y = r\sin\theta$

$\Rightarrow r^2 = x^2 + y^2$, $\theta = \tan^{-1}\frac{y}{x}$

i.e. r and θ are the function of x and y

$\therefore$ u is composite function of x and y

$$\therefore \frac{\partial u}{\partial x} = \frac{\partial u}{\partial r}\frac{\partial r}{\partial x} + \frac{\partial u}{\partial \theta}\cdot\frac{\partial \theta}{\partial x} \quad \text{---(1)}$$

$$\& \quad \frac{\partial u}{\partial y} = \frac{\partial u}{\partial r}\cdot\frac{\partial r}{\partial y} + \frac{\partial u}{\partial \theta}\cdot\frac{\partial \theta}{\partial y} \quad \text{---(2)}$$

Here $r^2 = x^2 + y^2$

$\Rightarrow 2r\frac{\partial r}{\partial x} = 2x$

$\Rightarrow \frac{\partial r}{\partial x} = \frac{x}{r} = \cos\theta$

and $\frac{\partial r}{\partial y} = \frac{y}{r} = \sin\theta$

$\because \theta = \tan^{-1} y/x$

$$\frac{\partial \theta}{\partial x} = \frac{1}{1+(\frac{y}{x})^2}\cdot\left(-\frac{1}{x^2}\right)$$

$$= \frac{-1}{x^2+y^2} = \frac{-\sin\theta}{r}$$

$$= -\frac{r\sin\theta}{r^2}$$

$$= -\frac{r\sin\theta}{r}$$

$\&$ $\frac{\partial \theta}{\partial y} = \frac{\cos\theta}{r}$

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$$\textcircled{1} \Rightarrow \frac{\partial u}{\partial x} = \cos\theta \frac{\partial u}{\partial r} - \frac{\sin\theta}{r} \frac{\partial u}{\partial \theta}$$

$$\therefore \frac{\partial}{\partial x} = \cos\theta \frac{\partial}{\partial r} - \frac{\sin\theta}{r} \frac{\partial}{\partial \theta}$$

$$\frac{\partial^2 u}{\partial x^2} = \frac{\partial}{\partial x}\left(\frac{\partial u}{\partial x}\right)$$

$$= \left(\cos\theta \frac{\partial}{\partial r} - \frac{\sin\theta}{r} \frac{\partial}{\partial \theta}\right)\left[\cos\theta \frac{\partial u}{\partial r} - \frac{\sin\theta}{r} \frac{\partial u}{\partial \theta}\right]$$

$$= \cos\theta \frac{\partial}{\partial r}\left[\cos\theta \frac{\partial u}{\partial r}\right] - \cos\theta \frac{\partial}{\partial r}\left[\frac{\sin\theta}{r} \frac{\partial u}{\partial \theta}\right]$$

$$- \frac{\sin\theta}{r} \frac{\partial}{\partial \theta}\left[\cos\theta \frac{\partial u}{\partial r}\right] + \frac{\sin\theta}{r} \frac{\partial}{\partial \theta}\left[\frac{\sin\theta}{r} \frac{\partial u}{\partial \theta}\right]$$

$$= \cos^2\theta \frac{\partial^2 u}{\partial r^2} - -\cos\theta \sin\theta \cdot \frac{\partial}{\partial r}\left(\frac{1}{r} \frac{\partial u}{\partial \theta}\right)$$

$$- \frac{\sin\theta}{r}\left[-\sin\theta \frac{\partial u}{\partial r} + \cos\theta \frac{\partial^2 u}{\partial r \partial \theta}\right]$$

$$+ \frac{\sin\theta}{r} \frac{1}{r}\left[\cos\theta \frac{\partial u}{\partial \theta} + \sin\theta \frac{\partial^2 u}{\partial \theta^2}\right]$$

$$= \cos^2\theta \frac{\partial^2 u}{\partial r^2} - \cos\theta \sin\theta \left[\frac{-1}{r^2} \frac{\partial u}{\partial \theta} + \frac{1}{r} \frac{\partial^2 u}{\partial r \partial \theta}\right]$$

$$+ \frac{\sin^2\theta}{r} \frac{\partial u}{\partial r} + \frac{\sin\theta\cos\theta}{r} \frac{\partial^2 u}{\partial r \partial \theta}$$

$$+ \frac{\sin^2\theta\cos\theta}{r^2} \frac{\partial u}{\partial \theta} + \frac{\sin^2\theta}{r^2} \frac{\partial^2 u}{\partial \theta^2}$$

$$\frac{\partial^2 u}{\partial x^2} = \cos^2\theta \frac{\partial^2 u}{\partial r^2} + \frac{\sin\theta\cos\theta}{r^2} \frac{\partial u}{\partial \theta} - \frac{\sin\theta\cos\theta}{r} \frac{\partial^2 u}{\partial r \partial \theta}$$

$$+ \frac{\sin^2\theta}{r} \frac{\partial u}{\partial r} - \frac{\sin\theta\cos\theta}{r} \frac{\partial^2 u}{\partial \theta \partial r} + \frac{\sin\theta\cos\theta}{r^2} \frac{\partial u}{\partial \theta}$$

$$+ \frac{\sin^2\theta}{r^2} \frac{\partial^2 u}{\partial \theta^2} \quad \textcircled{2}$$

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$$\Rightarrow \frac{\partial^2 u}{\partial x^2} = \cos^2\theta \frac{\partial^2 u}{\partial r^2} + 2\frac{\sin\theta\cos\theta}{r}\frac{\partial u}{\partial \theta} - 2\frac{\sin\theta\cos\theta}{r^2}\frac{\partial u}{\partial \theta} + \frac{\sin^2\theta}{r}\frac{\partial u}{\partial r} + \frac{\sin^2\theta}{r^2}\frac{\partial^2 u}{\partial \theta^2} \quad \text{---(3)}$$

Now from (2).

$$\frac{\partial u}{\partial y} = \frac{\partial u}{\partial r}\cdot\frac{\partial r}{\partial y} + \frac{\partial u}{\partial \theta}\frac{\partial \theta}{\partial y}$$

$$= \sin\theta \frac{\partial u}{\partial r} + \frac{\cos\theta}{r}\frac{\partial u}{\partial \theta}$$

$$\therefore \Rightarrow \frac{\partial}{\partial y} = \left(\sin\theta \frac{\partial}{\partial r} + \frac{\cos\theta}{r}\frac{\partial}{\partial \theta}\right)$$

$$\therefore \frac{\partial^2 u}{\partial y^2} = \frac{\partial}{\partial y}\left(\frac{\partial u}{\partial y}\right)$$

$$= \left(\sin\theta \frac{\partial}{\partial r} + \frac{\cos\theta}{r}\frac{\partial}{\partial \theta}\right)\left(\sin\theta \frac{\partial u}{\partial r} + \frac{\cos\theta}{r}\frac{\partial u}{\partial \theta}\right)$$

$$= \sin^2\theta \frac{\partial^2 u}{\partial r^2} + \frac{2\sin\theta\cos\theta}{r}\frac{\partial^2 u}{\partial r\partial\theta} + \frac{\cos^2\theta}{r^2}\frac{\partial^2 u}{\partial\theta^2} + \frac{\cos^2\theta}{r}\frac{\partial u}{\partial r} - \frac{2\sin\theta\cos\theta}{r^2}\frac{\partial u}{\partial\theta} \quad \text{---(4)}$$

adding eqn (3) and (4) we get.

$$\left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}\right) = \frac{\partial^2 u}{\partial r^2} + \frac{1}{r^2}\frac{\partial^2 u}{\partial\theta^2} + \frac{1}{r}\frac{\partial u}{\partial r}$$

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Ex: If $z = f(x,y)$, where, $x = r\cos\theta$, $y = r\sin\theta$ then prove that

$$\left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2 = \left(\frac{\partial z}{\partial r}\right)^2 + \frac{1}{r^2}\left(\frac{\partial z}{\partial \theta}\right)^2$$

Soln: Let $x = r\cos\theta$, $y = r\sin\theta$

$$\Rightarrow r^2 = x^2 + y^2 \quad \text{——— (1)}$$

and $$\theta = \tan^{-1}\frac{y}{x} \quad \text{——— (2)}$$

Now, let z is function of r and θ ans. whereas r and θ both are function of x and y

$\therefore z$ is composite function of x and y

$$\therefore \frac{\partial z}{\partial x} = \frac{\partial z}{\partial r}\frac{\partial r}{\partial x} + \frac{\partial z}{\partial \theta}\frac{\partial \theta}{\partial x} \quad \text{——— (3)}$$

$$\& \quad \frac{\partial z}{\partial y} = \frac{\partial z}{\partial r}\frac{\partial r}{\partial y} + \frac{\partial z}{\partial \theta}\frac{\partial \theta}{\partial y} \quad \text{——— (4)}$$

Now. Here Diff eqn (1) and (2) partially w.r.to. x and θ, we get

$$2r\frac{\partial r}{\partial x} = 2x \Rightarrow \frac{\partial r}{\partial x} = \frac{x}{r} = \frac{r\cos\theta}{r} = \cos\theta$$

$$2r\frac{\partial r}{\partial y} = 2y \Rightarrow \frac{\partial r}{\partial y} = \frac{y}{r} = \sin\theta$$

$$\frac{\partial \theta}{\partial x} = \frac{1}{1+\left(\frac{y}{x}\right)^2}\cdot y\left(\frac{-1}{x^2}\right) = \frac{-y}{x^2+y^2} = \frac{-r\sin\theta}{r^2} = -\frac{\sin\theta}{r}$$

$$\& \quad \frac{\partial \theta}{\partial y} = \frac{1}{1+\left(\frac{y}{x}\right)^2}\cdot\left(\frac{1}{x}\right) = \frac{x}{x^2+y^2} = \frac{r\cos\theta}{r^2} = \frac{\cos\theta}{r}$$

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Substituting these value in eqn (2) and (3) we get

$$\frac{\partial z}{\partial \xi} = \frac{\partial z}{\partial r}\cdot\frac{\partial r}{\partial \xi} + \frac{\partial z}{\partial \theta}\cdot\frac{\partial \theta}{\partial \xi}$$

$$\Rightarrow \frac{\partial z}{\partial \xi} = \cos\theta \frac{\partial z}{\partial r} + \frac{\sin\theta}{r}\frac{\partial z}{\partial \theta} \quad \text{---(A)}$$

and $$\frac{\partial z}{\partial \eta} = \frac{\partial z}{\partial r}\cdot\frac{\partial r}{\partial \eta} + \frac{\partial z}{\partial \theta}\frac{\partial \theta}{\partial \eta}$$

$$\Rightarrow \frac{\partial z}{\partial \eta} = \sin\theta \frac{\partial z}{\partial r} + \frac{r\cos\theta}{r}\frac{\partial z}{\partial \theta} \quad \text{---(B)}$$

Now on squaring and adding eqn (A) and (B) we get

$$\left(\frac{\partial z}{\partial \xi}\right)^2 + \left(\frac{\partial z}{\partial \eta}\right)^2 = \left(\cos\theta \frac{\partial z}{\partial r} + \frac{\sin\theta}{r}\frac{\partial z}{\partial \theta}\right)^2 + \left(\sin\theta \frac{\partial z}{\partial r} + \frac{r\cos\theta}{r}\frac{\partial z}{\partial \theta}\right)^2$$

$$= \cos^2\theta\cdot\left(\frac{\partial z}{\partial r}\right)^2 + 2\cos\theta\frac{\sin\theta}{r}\frac{\partial z}{\partial r}\frac{\partial z}{\partial \theta} + \frac{\sin^2\theta}{r^2}\left(\frac{\partial z}{\partial \theta}\right)^2$$

$$+ \sin^2\theta\left(\frac{\partial z}{\partial r}\right)^2 + 2\sin\theta\frac{\cos\theta}{r}\frac{\partial z}{\partial r}\frac{\partial z}{\partial \theta} + \frac{\cos^2\theta}{r^2}\left(\frac{\partial z}{\partial \theta}\right)^2$$

$$= (\cos^2\theta + \sin^2\theta)\left(\frac{\partial z}{\partial r}\right)^2 + \frac{(\sin^2\theta + \cos^2\theta)}{r^2}\left(\frac{\partial z}{\partial \theta}\right)^2$$

$$= \left(\frac{\partial z}{\partial r}\right)^2 + \frac{1}{r^2}\left(\frac{\partial z}{\partial \theta}\right)^2$$

$\because \cos^2\theta + \sin^2\theta = 1$

$$\therefore \left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2 = \left(\frac{\partial z}{\partial r}\right)^2 + \frac{1}{r^2}\left(\frac{\partial z}{\partial \theta}\right)^2$$

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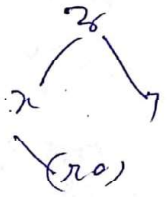
Let $z = f(x, y)$, $x = r\cos\theta$, $y = r\sin\theta$

Let $\left[r = \sqrt{x^2+y^2}, \quad \theta = \tan^{-1}(y/x)\right]$

Let z is compos



$$\frac{\partial z}{\partial r} = \frac{\partial z}{\partial x}\cdot\frac{\partial x}{\partial r} + \frac{\partial z}{\partial y}\cdot\frac{\partial y}{\partial r}$$

$$= \cos\theta \frac{\partial z}{\partial x} + \sin\theta \frac{\partial z}{\partial y} = \text{eqn}$$

$$\Rightarrow \left(\frac{\partial z}{\partial r}\right)^2 = \cos^2\theta \left(\frac{\partial z}{\partial x}\right)^2 + 2\sin\theta\cos\theta \frac{\partial z}{\partial x}\frac{\partial z}{\partial y} + \sin^2\theta\left(\frac{\partial z}{\partial y}\right)^2$$

and $$\frac{\partial z}{\partial \theta} = \frac{\partial z}{\partial x}\cdot\frac{\partial x}{\partial \theta} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial \theta}$$

$$= -r\sin\theta \frac{\partial z}{\partial x} + r\cos\theta \frac{\partial z}{\partial y}$$

$$\therefore \left(\frac{\partial z}{\partial \theta}\right)^2 = r^2\sin^2\theta\left(\frac{\partial z}{\partial x}\right)^2 - 2r^2\sin\theta\cos\theta \frac{\partial z}{\partial x}\frac{\partial z}{\partial y} + r^2\cos^2\theta\left(\frac{\partial z}{\partial y}\right)^2$$

$$\Rightarrow \frac{1}{r^2}\left(\frac{\partial z}{\partial \theta}\right)^2 = \sin^2\theta\left(\frac{\partial z}{\partial x}\right)^2 - 2\sin\theta\cos\theta + \cos^2\theta\left(\frac{\partial z}{\partial y}\right)^2$$

$$\boxed{\left(\frac{\partial z}{\partial r}\right)^2 + \frac{1}{r^2}\left(\frac{\partial z}{\partial \theta}\right)^2 = \left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2}$$

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Jacobian

If u and v are function of two independent variable x and y then the determinant

$$\begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{vmatrix}$$ is called the Jacobian of u, v w.r.to. x, y and it is denoted by

$$\frac{\partial(u,v)}{\partial(x,y)} \text{ or } J\left(\frac{uv}{x,y}\right).$$

Similarly, if $\frac{\partial(u,v,w)}{\partial(x,y,z)} = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} & \frac{\partial u}{\partial z} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} & \frac{\partial v}{\partial z} \\ \frac{\partial w}{\partial x} & \frac{\partial w}{\partial y} & \frac{\partial w}{\partial z} \end{vmatrix}$

Ie: ① Sometime Jacobian are known as functional determinant

② The Jacobian are useful in obtaining the partial derivatives of implicit function:

$f(x,y,u,v) = 0$ and $g(x,y,u,v) = 0$, these functional relation express u, v as functions of independent variable x, y then

$$\frac{\partial u}{\partial x} = -\frac{\frac{\partial(f,y)}{\partial(x,v)}}{\frac{\partial(f,g)}{\partial(u,v)}}; \quad \frac{\partial u}{\partial y} = -\frac{\frac{\partial(f,g)}{\partial(y,v)}}{\frac{\partial(f,g)}{\partial(u,v)}}$$

$$\frac{\partial u}{\partial x} = -\frac{\frac{\partial(f,g)}{\partial(u,x)}}{\frac{\partial(f,g)}{\partial(u,v)}}; \quad \frac{\partial v}{\partial y} = -\frac{\frac{\partial(f,g)}{\partial(u,y)}}{\frac{\partial(f,g)}{\partial(u,v)}}$$

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properties of Jacobians

(1) If u and v are functions of x and y then

$$J \cdot J^* = 1$$

where, $J = \frac{\partial(u,v)}{\partial(x,y)}$ and $J^* = \frac{\partial(x,y)}{\partial(u,v)}$

proof :⇒ Let u and v are two functions of x and y

$$u = f(x,y) \text{ and } v = g(x,y) \quad \text{---(1)}$$

writing x and y in terms of u and v

$$x = \phi_1(u,v) \text{ and } y = \phi_2(u,v) \quad \text{---(2)}$$

Differentiating eqn (1) partially w.r.to. u and v.

$$\frac{\partial u}{\partial u} = 1 = \frac{\partial u}{\partial x}\cdot\frac{\partial x}{\partial u} + \frac{\partial u}{\partial y}\cdot\frac{\partial y}{\partial u} \quad \text{---(3)}$$

$$\frac{\partial u}{\partial v} = 0 = \frac{\partial u}{\partial x}\cdot\frac{\partial x}{\partial v} + \frac{\partial u}{\partial y}\cdot\frac{\partial y}{\partial v} \quad \text{---(4)}$$

$$\frac{\partial v}{\partial u} = 0 = \frac{\partial v}{\partial x}\cdot\frac{\partial x}{\partial u} + \frac{\partial v}{\partial y}\cdot\frac{\partial y}{\partial u} \quad \text{---(5)}$$

$$\frac{\partial v}{\partial v} = 1 = \frac{\partial v}{\partial x}\frac{\partial x}{\partial v} + \frac{\partial v}{\partial y}\cdot\frac{\partial y}{\partial v} \quad \text{---(6)}$$

Now, $J \times J^* = \frac{\partial(u,v)}{\partial(x,y)} \times \frac{\partial(x,y)}{\partial(u,v)}$

$$= \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{vmatrix} \times \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix}$$

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$$= \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{vmatrix} \times \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix}$$

(∵ Interchanging rows and columns of second determinant)

$$= \begin{vmatrix} \frac{\partial u}{\partial x}\cdot\frac{\partial x}{\partial u} + \frac{\partial u}{\partial y}\cdot\frac{\partial y}{\partial u} & , \quad \frac{\partial u}{\partial x}\cdot\frac{\partial x}{\partial v} + \frac{\partial u}{\partial y}\cdot\frac{\partial y}{\partial v} \\ \frac{\partial v}{\partial x}\cdot\frac{\partial x}{\partial u} + \frac{\partial v}{\partial y}\cdot\frac{\partial y}{\partial u} & \frac{\partial v}{\partial x}\cdot\frac{\partial x}{\partial v} + \frac{\partial v}{\partial y}\cdot\frac{\partial y}{\partial v} \end{vmatrix}$$

$$= \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix}$$

(∵ by eqn ③ ④ ⑤ and ⑥)

$$= 1$$

$$\therefore \boxed{J \times J^* = 1}$$

Hence proved.

Similarly

$$J \times J^* = \frac{\partial(u,v,w)}{\partial(x,y,z)} \times \frac{\partial(x,y,z)}{\partial(u,v,w)} = 1$$

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(II) If functions u, v, w of ~~three~~ two independent variables x, y, z are not independent (dependents) then

$$\frac{\partial(u,v,w)}{\partial(x,y,z)} = 0$$

Proof: If u, v, w are not independent (dependent) then there must be functional relation betn u, v

let $f(u,v) = 0$, be the relation betn u and v ——— (1)

Differentiating this relation w.r.to. x, y we get

$$\frac{\partial f}{\partial u}\cdot\frac{\partial u}{\partial x} + \frac{\partial f}{\partial v}\cdot\frac{\partial v}{\partial x} = 0 \quad ——— (2)$$

& $$\frac{\partial f}{\partial u}\cdot\frac{\partial u}{\partial y} + \frac{\partial f}{\partial v}\cdot\frac{\partial v}{\partial y} = 0 \quad ——— (3)$$

$$\Rightarrow \begin{vmatrix} \frac{\partial f}{\partial u} & \frac{\partial u}{\partial x} & \frac{\partial v}{\partial x} \\ \frac{\partial f}{\partial u} & \frac{\partial u}{\partial y} & \frac{\partial v}{\partial y} \end{vmatrix} = 0$$

Eliminating $\frac{\partial f}{\partial u}$, and $\frac{\partial f}{\partial v}$ from eqn (2) and (3) we get

$$\left(\frac{\partial f}{\partial u}\cdot\frac{\partial u}{\partial x}\right)\begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial v}{\partial x} \\ \frac{\partial u}{\partial y} & \frac{\partial v}{\partial y} \end{vmatrix} = 0$$

$$\Rightarrow \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{vmatrix} = 0 \quad \left(\text{interchanging rows and columns}\right)$$

$$\Rightarrow \boxed{\frac{\partial(u,v)}{\partial(x,y)} = 0}$$

Ily If ~~u,v,w~~ are function u, v, w of ~~three~~ independent v
x, y, z are not independent then

$$\frac{\partial(u,v,w)}{\partial(x,y,z)} = 0$$

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(III) If u, v, are functions of r, s and r, s are different functions of x, y then

$$\frac{\partial(u,v)}{\partial(x,y)} = \frac{\partial(u,v)}{\partial(r,s)} \times \frac{\partial(r,s)}{\partial(x,y)}$$

Proof: Since u, v are function of r, s

$$\therefore \frac{\partial(u,v)}{\partial(r,s)} = \begin{vmatrix} \frac{\partial u}{\partial r} & \frac{\partial u}{\partial s} \\ \frac{\partial v}{\partial r} & \frac{\partial v}{\partial s} \end{vmatrix} \quad \text{---(1)}$$

and since r, s are function of x, y.

$$\therefore \frac{\partial(r,s)}{\partial(x,y)} = \begin{vmatrix} \frac{\partial r}{\partial x} & \frac{\partial r}{\partial y} \\ \frac{\partial s}{\partial x} & \frac{\partial s}{\partial y} \end{vmatrix} \quad \text{---(2)}$$

Now

$$\frac{\partial(u,v)}{\partial(r,s)} \times \frac{\partial(r,s)}{\partial(x,y)} = \begin{vmatrix} \frac{\partial u}{\partial r} & \frac{\partial u}{\partial s} \\ \frac{\partial v}{\partial r} & \frac{\partial v}{\partial s} \end{vmatrix} \times \begin{vmatrix} \frac{\partial r}{\partial x} & \frac{\partial r}{\partial y} \\ \frac{\partial s}{\partial x} & \frac{\partial s}{\partial y} \end{vmatrix}$$

$$= \begin{vmatrix} \frac{\partial u}{\partial r} & \frac{\partial u}{\partial s} \\ \frac{\partial v}{\partial r} & \frac{\partial v}{\partial s} \end{vmatrix} \times \begin{vmatrix} \frac{\partial r}{\partial x} & \frac{\partial s}{\partial x} \\ \frac{\partial r}{\partial y} & \frac{\partial s}{\partial y} \end{vmatrix}$$ (∵ interchanging row and column of second determinant)

$$= \begin{vmatrix} \frac{\partial u}{\partial r}\frac{\partial r}{\partial x} + \frac{\partial u}{\partial s}\frac{\partial s}{\partial x} & \frac{\partial u}{\partial r}\frac{\partial r}{\partial y} + \frac{\partial u}{\partial s}\frac{\partial s}{\partial y} \\ \frac{\partial v}{\partial r}\frac{\partial r}{\partial x} + \frac{\partial v}{\partial s}\frac{\partial s}{\partial x} & \frac{\partial v}{\partial r}\frac{\partial r}{\partial y} + \frac{\partial v}{\partial s}\frac{\partial s}{\partial y} \end{vmatrix}$$

$$= \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{vmatrix} = \frac{\partial(u,v)}{\partial(x,y)}$$

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Ex: If $x = u(1-v)$ and $y = uv$.

prove that. $J \cdot J' = 1$

where $J = \frac{\partial(x,y)}{\partial(u,v)}$

Soln: let $x = u(1-v)$

& $y = uv$

Then, $\frac{\partial x}{\partial u} = (1-v)$, $\frac{\partial x}{\partial v} = -u$

$\frac{\partial y}{\partial u} = v$, $\frac{\partial y}{\partial v} = u$

$\therefore J = \frac{\partial(x,y)}{\partial(u,v)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix} = \begin{vmatrix} (1-v) & -u \\ v & u \end{vmatrix}$

$= (1-v)\cdot u + u \cdot v$

$= u - vu + vu$

$\Rightarrow \boxed{J = \frac{\partial(x,y)}{\partial(u,v)} = u}$ ——— (1)

Now, $x = u(1-v) = u - uv$

but $y = uv$

$\therefore x = u - y \Rightarrow u = x + y$

and $y = uv \Rightarrow v = \frac{y}{u} = \frac{y}{x+y}$

$\therefore \frac{\partial u}{\partial x} = 1$, $\frac{\partial u}{\partial y} = 1$

$\frac{\partial v}{\partial x} = \frac{-y}{(x+y)^2}$, $\frac{\partial v}{\partial y} = \frac{(x+y) - y}{(x+y)^2} = \frac{x}{(x+y)^2}$

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$$\therefore J' = \frac{\partial(u,v)}{\partial(x,y)} = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{vmatrix}$$

$$= \begin{vmatrix} 1 & 1 \\ \frac{-y}{(x+y)^2} & \frac{x}{(x+y)^2} \end{vmatrix}$$

$$\boxed{J' = \frac{1}{x+y} = \frac{1}{4}} \quad \text{---------- (2)}$$

from eqn (1) and (2)

$$J \cdot J' = 4 \cdot \frac{1}{4} = 1$$

$$\therefore \boxed{J \cdot J' = 1}$$ proved

Ex: If $x+y = 2e^{\theta}\cos\phi$ and $x-y = 2ie^{\theta}\sin\phi$

verify that $\frac{\partial(x,y)}{\partial(\theta,\phi)} \times \frac{\partial(\theta,\phi)}{\partial(x,y)} = 1$

Soln: ⇒ Let $x+y = 2e^{\theta}\cos\phi$ ---------- (1)

& $x-y = 2ie^{\theta}\sin\phi$ ---------- (2)

adding eqn (1) and (2) we get

$$2x = 2e^{\theta}(\cos\phi + i\sin\phi)$$

$$\Rightarrow x = e^{\theta} \cdot e^{i\phi} = e^{\theta+i\phi} \quad \text{---------- (3)}$$

substracting eqn (1) and (2) we get

$$2y = 2e^{\theta}(\cos\phi - i\sin\phi) = 2e^{\theta}e^{-i\phi}$$

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$$\Rightarrow y = e^{\theta - i\phi} \quad \text{---(4)}$$

Differentiating eqn (3) and (4) partially w.r.to θ and ϕ we get

$$\frac{\partial x}{\partial \theta} = e^{\theta + i\phi}, \quad \frac{\partial x}{\partial \phi} = i e^{\theta + i\phi}$$

$$\frac{\partial y}{\partial \theta} = e^{\theta - i\phi}, \quad \frac{\partial y}{\partial \phi} = -i e^{\theta - i\phi}$$

$$\therefore \frac{\partial(x,y)}{\partial(\theta,\phi)} = \begin{vmatrix} \frac{\partial x}{\partial \theta} & \frac{\partial x}{\partial \phi} \\ \frac{\partial y}{\partial \theta} & \frac{\partial y}{\partial \phi} \end{vmatrix}$$

$$= \begin{vmatrix} e^{\theta + i\phi} & i e^{\theta + i\phi} \\ e^{\theta - i\phi} & -i e^{\theta - i\phi} \end{vmatrix}$$

$$= \left(-i e^{\theta + i\phi} \cdot e^{\theta - i\phi}\right) - \left(i e^{\theta + i\phi} e^{\theta - i\phi}\right)$$

$$= -i e^{2\theta} - i e^{2\theta}$$

$$\boxed{\frac{\partial(x,y)}{\partial(\theta,\phi)} = -2i e^{2\theta}} \quad \text{---(5)}$$

Now, ~~Squaring~~ multiplying eqn (3) and (4) we get

$$x \cdot y = e^{\theta + i\phi} \cdot e^{\theta + i\phi}$$

$$xy = e^{2\theta}$$

$$2\theta = \log(x \cdot y)$$

$$\Rightarrow \theta = \frac{1}{2}\log(xy) \quad \text{---(6)}$$

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and dividing eqn (3) by (4) we get

$$\frac{x}{y} = \frac{e^{\theta+i\phi}}{e^{\theta-i\phi}}$$

$$= e^{\theta+i\phi} \cdot e^{-\theta+i\phi}$$

$$\frac{x}{y} = e^{2i\phi}$$

$$\Rightarrow 2i\phi = \log\left(\frac{x}{y}\right)$$

$$\Rightarrow \phi = \frac{1}{2i}\log\left(\frac{x}{y}\right)$$

$$\Rightarrow \phi = \frac{i}{2}\log\left(\frac{x}{y}\right) \quad \text{---(7)} \quad \because (-1 = i^2)$$

Diff eqn (6) and (7) partially w.r.to x, y. we get

$$\frac{\partial\theta}{\partial x} = \frac{1}{2}\frac{1}{xy}\cdot y = \frac{1}{2x}$$

$$\frac{\partial\theta}{\partial y} = \frac{1}{2}\frac{1}{xy}\, x = \frac{1}{2y}$$

$$\frac{\partial\phi}{\partial x} = \frac{i}{2}\left(\frac{y}{x}\right)\cdot\left(\frac{1}{y}\right) = \frac{i}{2x}$$

$$\frac{\partial\phi}{\partial y} = \frac{i}{2}\left(\frac{y}{x}\right)\cdot x\left(\frac{-1}{y^2}\right) = \frac{-i}{2y}$$

Thus

$$\frac{\partial(\theta,\phi)}{\partial(x,y)} = \begin{vmatrix} \frac{\partial\theta}{\partial x} & \frac{\partial\theta}{\partial y} \\ \frac{\partial\phi}{\partial x} & \frac{\partial\phi}{\partial y} \end{vmatrix} = \begin{vmatrix} \frac{1}{2x} & \frac{1}{2y} \\ \frac{i}{2x} & -\frac{i}{2y} \end{vmatrix}$$

$$= \begin{vmatrix} -\frac{i}{2xy} & -\frac{i}{4xy} \end{vmatrix} = -\frac{2i}{4xy}$$

$$= \frac{-i}{2(e^{\theta+i\phi}\, e^{\theta+i\phi})} = \frac{-i}{2e^{2\theta}} \quad \text{---(8)}$$

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from eqn (6) and (8) we get

$$\frac{\partial(x,y)}{\partial(\theta,\phi)} \times \frac{\partial(\theta,\phi)}{\partial(x,y)} = (-2ie^{2\theta}) \cdot \left(\frac{-i}{2e^{2\theta}}\right)$$

$$= (-i)(-i) = -i^2$$

$$= -(-1) = 1$$

$$\Rightarrow \boxed{\frac{\partial(x,y)}{\partial(\theta,\phi)} \times \frac{\partial(\theta,\phi)}{\partial(x,y)} = 1}$$

Ex: If $x = a\cosh\xi\cos\eta$, $y = a\sinh\xi\sin\eta$

Then prove. $\frac{\partial(\xi,\eta)}{\partial(x,y)} = \frac{2}{a^2} \cdot \frac{1}{(\cosh 2\xi - \cos 2\eta)}$

Soln:⇒ We know that.

$$J \cdot J' = \frac{\partial(x,y)}{\partial(\xi,\eta)} \times \frac{\partial(\xi,\eta)}{\partial(x,y)} = 1 \quad \text{——(1)}$$

Now. $x = a\cosh\xi\cos\eta$ ——(2)

$y = a\sinh\xi\sin\eta$ ——(3)

Diff eqn (2) and (3) partially w.r to. ξ and η w

$$\frac{\partial x}{\partial \xi} = a\sinh\xi\cos\eta$$

$$\frac{\partial x}{\partial \eta} = -a\cosh\xi\sin\eta$$

$$\frac{\partial y}{\partial \xi} = a\cosh\xi\sin\eta$$

$$\frac{\partial y}{\partial \eta} = a\sinh\xi\cos\eta.$$

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$$\frac{\partial(x,y)}{\partial(\xi,\eta)} = \begin{vmatrix} \frac{\partial x}{\partial \xi} & \frac{\partial x}{\partial \eta} \\ \frac{\partial y}{\partial \xi} & \frac{\partial y}{\partial \eta} \end{vmatrix}$$

$$= \begin{vmatrix} a\sinh\xi\cos\eta & -a\cosh\xi\sin\eta \\ a\cosh\xi\sin\eta & a\sinh\xi\cos\eta \end{vmatrix}$$

$$= a^2\sinh^2\xi\cos^2\eta + a^2\cosh^2\xi\sin^2\eta$$

$$= a^2\left[\sinh^2\xi\cos^2\eta + \cosh^2\xi\sin^2\eta\right]$$

$$= a^2\left[\sinh^2\xi\cdot\left(\frac{1+\cos 2\eta}{2}\right) + \cosh^2\xi\left(\frac{1-\cos 2\eta}{2}\right)\right]$$

$\because \boxed{\cos 2\theta = 2\cos^2\theta - 1 = 1 - 2\sin^2\theta}$

$$= \frac{a^2}{2}\left[\sinh^2\xi + \sinh^2\xi\cdot\cos 2\eta + \cosh^2\xi - \cosh^2\xi\cos 2\eta\right]$$

$$= \frac{a^2}{2}\left[\sinh^2\xi + \cosh^2\xi - \cos 2\eta\left(\cosh^2\xi - \sinh^2\xi\right)\right]$$

$$= \frac{a^2}{2}\left[\sinh^2\xi + \cosh^2\xi - \cos 2\eta\ (1)\right]$$

$$\frac{\partial(x,y)}{\partial(\xi,\eta)} = \frac{a^2}{2}\left[\cosh 2\xi - \cos 2\eta\right] \quad \because \boxed{\begin{aligned}&\cosh^2\xi - \sinh^2\xi = 1 \\ &\cosh^2\xi + \sinh^2\xi = \cosh 2\xi\end{aligned}}$$

$\Rightarrow$ by using (1)

$$(1) \Rightarrow \frac{a^2}{2}\left(\cosh 2\xi - \cos 2\eta\right)\cdot\frac{\partial(\xi,\eta)}{\partial(x,y)} = 1$$

$$\Rightarrow \boxed{\frac{\partial(\xi,\eta)}{\partial(x,y)} = \frac{1}{a^2/2\left(\cosh 2\xi - \cos 2\eta\right)}}$$

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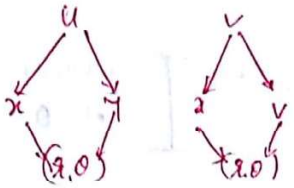
Ex: If $u = x^2 - y^2$, $v = 2xy$ where $x = r\cos\theta$, & $y = r\sin\theta$. find $\frac{\partial(u,v)}{\partial(r,\theta)}$.

Soln: let $u = x^2 - y^2$ —— (1)

& $v = 2xy$ —— (2)



Differentiating eqn (1) and (2) we get

$\frac{\partial u}{\partial x} = 2x, \quad \frac{\partial u}{\partial y} = -2y$

$\frac{\partial v}{\partial x} = 2y \quad \frac{\partial v}{\partial y} = 2x$

$\therefore \frac{\partial(u,v)}{\partial(x,y)} = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{vmatrix} = \begin{vmatrix} 2x & -2y \\ 2y & 2x \end{vmatrix}$

$= 4(x^2+y^2) = 4r^2$ —— (3)

Now let $x = r\cos\theta$, —— (4)

& $y = r\sin\theta$ —— (5)

Differentiating (4) and (5) w.r.to. r, θ we get

$\frac{\partial x}{\partial r} = \cos\theta, \quad \frac{\partial x}{\partial \theta} = -r\sin\theta$

$\frac{\partial y}{\partial r} = \sin\theta \quad \frac{\partial y}{\partial \theta} = r\cos\theta$

$\therefore \frac{\partial(x,y)}{\partial(r,\theta)} = \begin{vmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} \end{vmatrix} = \begin{vmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{vmatrix}$

$= r\cos^2\theta + r\sin^2\theta$

$= r(\cos^2\theta + \sin^2\theta)$

$= r$ —— (6)

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Now we know that:

$$\frac{\partial(u,v)}{\partial(r,\theta)} = \frac{\partial(u,v)}{\partial(x,y)} \cdot \frac{\partial(x,y)}{\partial(r,\theta)}$$

$$= 4r^2 \cdot r$$

$$= 4r^3$$

or $$\boxed{\frac{\partial(u,v)}{\partial(r,\theta)} = 4(x^2+y^2)^3}$$

Ex:⇒ If $u = e^x \cos y$, $v = e^x \sin y$, where $x = lr + sm$ and $y = mr - ls$ verify chain rule of Jacobian. l, m being constan

Soln:⇒ let $u = e^x \cos y$ ——— (1)

& $v = e^x \sin y$ ——— (2)

Differentiating eqn (1) & (2) partially w.r.to. x and y.

$$\frac{\partial u}{\partial x} = e^x \cos y \qquad \frac{\partial u}{\partial y} = -e^x \sin y$$

$$\frac{\partial v}{\partial x} = e^x \sin y \qquad \frac{\partial v}{\partial y} = e^x \cos y$$

$$\therefore \left| \frac{\partial(u,v)}{\partial(x,y)} \right| = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{vmatrix} = \begin{vmatrix} e^x \cos y & -e^x \sin y \\ e^x \sin y & e^x \cos y \end{vmatrix}$$

$$= e^{2x} \cos^2 y + e^{2x} \sin^2 y = e^{2x}(1$$

$$= e^{2x} \qquad \text{——— (3)}$$

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Now. $x = lr + sm$ ——— (4) and $y = mr - ls$. ——— (5)

where l, m are const.

Differentiating these eqn's partially w.r.to. r and s.

$$\frac{\partial x}{\partial r} = l \qquad \frac{\partial x}{\partial s} = m$$

$$\frac{\partial y}{\partial r} = m \qquad \frac{\partial y}{\partial s} = -l$$

$$\therefore \frac{\partial(x,y)}{\partial(r,s)} = \begin{vmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial s} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial s} \end{vmatrix} = \begin{vmatrix} l & m \\ m & -l \end{vmatrix}$$

$$= -(l^2 + m^2) \qquad \text{——— (6)}$$

from eqn (3) and (6).

$$\frac{\partial(u,v)}{\partial(x,y)} \times \frac{\partial(x,y)}{\partial(r,s)} = -e^{2x}(l^2+m^2) \qquad \text{——— (7)}$$

Ex: $u = e^x \cos y = e^{lr+ms} \cos(mr - ls)$

& $v = e^x \sin y = e^{lr+ms} \sin(mr - ls)$

Differentiating these eqn's partially w.r.to. r and s.

$$\frac{\partial u}{\partial r} = \left[l e^{lr+ms} \cdot \cos(mr - ls) + m e^{lr+ms} \sin(mr - ls) \right]$$

$$= l e^x \cos y - m e^x \sin y$$

$$\frac{\partial u}{\partial s} = m e^{lr+ms} \cos(mr - ls) + l e^{lr+ms} \sin(mr - sl)$$

$$= m e^x \cos y + l e^x \sin y.$$

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$$\frac{\partial v}{\partial r} = l e^{lr+ms} \sin(mr-ls) + m e^{lr+ms} \cos(mr-ls)$$

$$= l e^x \sin y + m e^x \cos y$$

$$\frac{\partial v}{\partial s} = m e^{lr+ms} \sin(mr-ls) - l e^{lr+ms} \cos(mr-ls)$$

$$= m e^x \sin y - l e^x \cos y$$

$$\therefore \frac{\partial(u,v)}{\partial(r,s)} = \begin{vmatrix} \frac{\partial u}{\partial r} & \frac{\partial u}{\partial s} \\ \frac{\partial v}{\partial r} & \frac{\partial v}{\partial s} \end{vmatrix}$$

$$= \begin{vmatrix} l e^x \cos y - m e^x \sin y & m e^x \cos y + l e^x \sin y \\ l e^x \sin y + m e^x \cos y & m e^x \sin y - l e^x \cos y \end{vmatrix}$$

$$= \begin{vmatrix} e^x (l \cos y - m \sin y) & e^x (m \cos y + l \sin y) \\ e^x (l \sin y + m \cos y) & e^x (m \sin y - l \cos y) \end{vmatrix}$$

$$= e^{2x} (l \cos y - m \sin y)(m \sin y - l \cos y) - e^{2x} (m \cos y + l \sin y)$$
$$(l \sin y + m \cos$$

$$= e^{2x} [lm \cos y \sin y - l^2 \cos^2 y - m^2 \sin^2 y + lm \sin y \cos y$$
$$- lm \cos y \sin y - m^2 \cos^2 y - l^2 \sin^2 y - lm \sin y \cos$$

$$= -e^{2x} (l^2 + m^2)(\sin^2 y + \cos^2 y)$$

$$\frac{\partial(u,v)}{\partial(r,s)} = -e^{2x} (l^2 + m^2) \quad \longleftarrow (8)$$

From eqn (7) and (8), we

$$\boxed{\frac{\partial(u,v)}{\partial(x,y)} \times \frac{\partial(x,y)}{\partial(r,s)} = \frac{\partial(u,v)}{\partial(r,s)}}$$

Hence Chain rule
Jacobian is verified

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Ex: If $x = \sqrt{vw}$, $y = \sqrt{uw}$, $z = \sqrt{uv}$ and $u = r\sin\theta\cos\phi$, $v = r\sin\theta\sin\phi$, $w = r\cos\theta$ find $\frac{\partial(x,y,z)}{\partial(r,\theta,\phi)}$

Sol: Let $x = \sqrt{vw}$, $y = \sqrt{uw}$, $z = \sqrt{uv}$

$\Rightarrow \frac{\partial x}{\partial u} = 0$, $\frac{\partial y}{\partial u} = \frac{1}{2}\sqrt{\frac{w}{u}}$, $\frac{\partial z}{\partial u} = \frac{1}{2}\sqrt{\frac{v}{u}}$

$\frac{\partial x}{\partial v} = \frac{1}{2\sqrt{vw}}\cdot w = \frac{1}{2}\sqrt{\frac{w}{v}}$, $\frac{\partial y}{\partial v} = 0$, $\frac{\partial z}{\partial v} = \frac{1}{2}\sqrt{\frac{u}{v}}$

$\frac{\partial x}{\partial w} = \frac{1}{2}\sqrt{\frac{v}{w}}$, $\frac{\partial y}{\partial w} = \frac{1}{2}\sqrt{\frac{u}{w}}$, $\frac{\partial z}{\partial w} = 0$

$$\therefore \frac{\partial(x,y,z)}{\partial(u,v,w)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} & \frac{\partial x}{\partial w} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} & \frac{\partial y}{\partial w} \\ \frac{\partial z}{\partial u} & \frac{\partial z}{\partial v} & \frac{\partial z}{\partial w} \end{vmatrix} = \begin{vmatrix} 0 & \frac{1}{2}\sqrt{\frac{w}{v}} & \frac{1}{2}\sqrt{\frac{v}{w}} \\ \frac{1}{2}\sqrt{\frac{w}{u}} & 0 & \frac{1}{2}\sqrt{\frac{u}{w}} \\ \frac{1}{2}\sqrt{\frac{u}{v}} & \frac{1}{2}\sqrt{\frac{v}{u}} & 0 \end{vmatrix}$$

$$= -\left[\frac{1}{2}\sqrt{\frac{w}{v}}\left(\frac{1}{4}\sqrt{\frac{w}{u}}\sqrt{\frac{u}{v}}\right) + \frac{1}{2}\sqrt{\frac{v}{w}}\left(\frac{1}{4}\sqrt{\frac{u}{w}}\sqrt{\frac{v}{u}}\right)\right]$$

$$= \frac{1}{8}\sqrt{\frac{w}{v}}\left(+\frac{1}{8}\sqrt{\frac{v}{w}}\right) + \frac{1}{8}\sqrt{\frac{v}{w}}\cdot\sqrt{\frac{w}{v}}$$

$$= \frac{1}{8} + \frac{1}{8} = \frac{2}{8} = \frac{1}{4}$$

If $u = r\sin\theta\cos\phi$, $v = r\sin\theta\sin\phi$, $w = r\cos\theta$

$\Rightarrow \frac{\partial u}{\partial r} = \sin\theta\cos\phi$, $\frac{\partial v}{\partial r} = \sin\theta\sin\phi$, $\frac{\partial w}{\partial r} = \cos\theta$

$\frac{\partial u}{\partial \theta} = r\cos\theta\cos\phi$, $\frac{\partial v}{\partial \theta} = r\cos\theta\sin\phi$, $\frac{\partial w}{\partial \theta} = -r\sin\theta$

$\frac{\partial u}{\partial \phi} = -r\sin\theta\sin\phi$, $\frac{\partial v}{\partial \phi} = r\sin\theta\cos\phi$, $\frac{\partial w}{\partial \phi} = 0$

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$$\therefore \frac{\partial(x,y,z)}{\partial(u,v,w)} = \begin{vmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} & \frac{\partial x}{\partial \phi} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} & \frac{\partial y}{\partial \phi} \\ \frac{\partial z}{\partial r} & \frac{\partial z}{\partial \theta} & \frac{\partial z}{\partial \phi} \end{vmatrix}$$

$$= \begin{vmatrix} \sin\theta\cos\phi & r\cos\theta\cos\phi & -r\sin\theta\sin\phi \\ \sin\theta\sin\phi & r\cos\theta\sin\phi & r\sin\theta\cos\phi \\ \cos\theta & -r\sin\theta & 0 \end{vmatrix}$$

$$= r \cdot r \begin{vmatrix} \sin\theta\cos\phi & \cos\theta\cos\phi & -\sin\theta\sin\phi \\ \sin\theta\sin\phi & \cos\theta\sin\phi & \sin\theta\cos\phi \\ \cos\theta & -\sin\theta & 0 \end{vmatrix}$$

$$= r^2 \left[\sin\theta\cos\phi \left(\sin^2\theta\cos\phi\right) - \cos\theta\cos\phi\left(-\sin\theta\cos\phi\right) \right.$$
$$\left. + \sin\theta\sin\phi\left(-\sin^2\theta\sin\phi - \cos^2\theta\right) \right]$$

$$= r^2 \left[\sin^3\theta\cos^2\phi + \sin\theta\cos^2\theta\cos^2\phi + \sin^3\theta\sin^2\phi + \cos^2\right.$$

$$= r^2 \left[\sin^3\theta\left(\cos^2\phi + \sin^2\phi\right) + \sin\theta\cos^2\theta\left(\cos^2\phi + \sin^2\right.\right.$$

$$= r^2 \left[\sin^3\theta + \sin\theta\cos^2\theta \right]$$

$$= r^2 \left[\sin\theta\left(\sin^2\theta + \cos^2\theta\right) \right]$$

$$\frac{\partial(u,v,w)}{\partial(r,\theta,\phi)} = r^2\sin\theta$$

$$\therefore \frac{\partial(x,y,z)}{\partial(r,\theta,\phi)} = \frac{\partial(x,y,z)}{\partial(u,v,w)} \times \frac{\partial(u,v,w)}{\partial(r,\theta,\phi)} = \frac{1}{4}\cdot r^2\sin\theta$$

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Ex: If $u = \frac{x+y}{x-y}$ and $v = \frac{xy}{(x-y)^2}$

find $\frac{\partial(u,v)}{\partial(x,y)}$, Are u and v functionally related? if so, find the relationship.

Soln: ⇒ Let $u = \frac{x+y}{x-y}$

$$\therefore \frac{\partial u}{\partial x} = \frac{(x-y)\cdot(1) - (x+y)\cdot(1)}{(x-y)^2} = \frac{-2y}{(x-y)^2}$$

& $$\frac{\partial u}{\partial y} = \frac{(x-y)(0) - (x+y)(-1)}{(x-y)^2} = \frac{2x}{(x-y)^2}$$

$$v = \frac{xy}{(x-y)^2}$$

$$\therefore \frac{\partial v}{\partial x} = \frac{(x-y)^2(y) - xy[2(x-y)\cdot]}{(x-y)^4} = \frac{(x-y)[(x-y)\cdot y - 2xy]}{(x-y)^{4-3}}$$

$$= \frac{xy - y^2 - 2xy}{(x-y)^3} = \frac{-[(xy) + y^2]}{(x-y)^3}$$

$$\frac{\partial v}{\partial y} = \frac{(x-y)^2\cdot(x) - (xy)\cdot 2(x-y)\cdot(-1)}{(x-y)^4}$$

$$= \frac{(x-y)[(x-y)(x) + 2xy]}{(x-y)^4} = \frac{x^2 - xy + 2xy}{(x-y)^3}$$

$$= \frac{x^2 + yx}{(x-y)^3}$$

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$$\therefore \frac{\partial(u,v)}{\partial(x,y)} = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{vmatrix}$$

$$= \begin{vmatrix} \frac{-2y}{(x-y)^2} & \frac{2x}{(x-y)^2} \\ \frac{-(xy+y^2)}{(x-y)^3} & \frac{(x^2+xy)}{(x-y)^3} \end{vmatrix}$$

$$= \frac{1}{(x-y)^2(x-y)^3} \begin{vmatrix} -2y & 2x \\ -(xy+y^2) & (x^2+xy) \end{vmatrix}$$

$$= \frac{-2y(x^2+xy) + 2x(xy+y^2)}{(x-y)^5}$$

$$= \frac{-2x^2y - 2xy^2 + 2x^2y + 2xy^2}{(x-y)^5} = \frac{0}{(x-y)^5}$$

$$\therefore \boxed{\frac{\partial(u,v)}{\partial(x,y)} = 0}$$

Thus u and v are functionally related.

Now we have $\boxed{u = \frac{x+y}{x-y}}$ $\quad \boxed{v = \frac{xy}{(x-y)^2}}$

$$\Rightarrow u^2 = \frac{(x+y)^2}{(x-y)^2} = \frac{x^2+y^2+2xy}{(x-y)^2} = \frac{(x-y)^2+2xy+2xy}{(x-y)^2}$$

$$= \frac{(x^2-2xy+y^2)+(4xy)}{(x-y)^2} = \frac{(x-y)^2+4xy}{(x-y)^2} = \frac{(x-y)^2}{(x-y)^2} + \frac{4xy}{(x-y)^2}$$

$$\Rightarrow \boxed{u^2 = 1+4v}$$

$\boxed{u = 1+4v}$ is the required relation.

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Ex: If $u = \sin^{-1}x + \sin^{-1}y$, $v = x\sqrt{1-y^2} + y\sqrt{1-x^2}$

find $\frac{\partial(u,v)}{\partial(x,y)}$. Also u and v are dependent functions. if so.. find their relations.

Soln: Let $u = \sin^{-1}x + \sin^{-1}y$

& $v = x\sqrt{1-y^2} + y\sqrt{1-x^2}$

Differentiating partially w.r.to x and y we get

$$\frac{\partial u}{\partial x} = \frac{1}{\sqrt{1-x^2}}, \quad \frac{\partial u}{\partial y} = \frac{1}{\sqrt{1-y^2}}$$

$$\frac{\partial v}{\partial x} = \sqrt{1-y^2} + \frac{-2xy}{2\sqrt{1-x^2}} = \sqrt{1-y^2} - \frac{xy}{\sqrt{1-x^2}}$$

$$\frac{\partial v}{\partial y} = \frac{-xy}{\sqrt{1-y^2}} + \sqrt{1-x^2}$$

$$\therefore \frac{\partial(u,v)}{\partial(x,y)} = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{vmatrix} = \begin{vmatrix} \frac{1}{\sqrt{1-x^2}} & \frac{1}{\sqrt{1-y^2}} \\ \sqrt{1-y^2} - \frac{xy}{\sqrt{1-x^2}} & \frac{-xy}{\sqrt{1-y^2}} + \sqrt{1-x^2} \end{vmatrix}$$

$$= \left(\frac{1}{\sqrt{1-x^2}}\right)\left(\frac{-xy}{\sqrt{1-y^2}} + \sqrt{1-x^2}\right) - \left(\frac{1}{\sqrt{1-y^2}}\right)\left(\sqrt{1-y^2} - \frac{xy}{\sqrt{1-x^2}}\right)$$

$$= \left(\frac{-xy}{\sqrt{1-x^2}\sqrt{1-y^2}} + 1\right) - \left(1 - \frac{xy}{\sqrt{1-y^2}\cdot\sqrt{1-x^2}}\right)$$

$$= 0$$

$\therefore \Rightarrow$ u and v are functionally dependent

Relation betn u and v

Let $\sin^{-1}x = \alpha \Rightarrow x = \sin\alpha$

$\sin^{-1}y = \beta \Rightarrow y = \sin\beta$

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$\therefore \boxed{u = \alpha + \beta}$

and $v = \sin\alpha\sqrt{1-\sin^2\beta} + \sin\beta\sqrt{1-\sin^2\alpha}$

$= \sin\alpha\cos\beta + \sin\beta\cos\alpha$

$= \sin(\alpha+\beta)$

$\boxed{v = \sin(u)}$

OR $\boxed{u = \sin^{-1}(v)}$ is the required relation.

Ex :⇒ If $u = x^2 e^{-y}\cosh z$, $v = x^2 e^{-y}\sinh z$, $w = 3x^4 e^{-2y}$

find $\frac{\partial(u,v,w)}{\partial(x,y,z)}$, are u, v, w functionally related?, if so find the relationship

Soln :⇒ Let $u = x^2 e^{-y}\cosh z$ ——— (1)

$v = x^2 e^{-y}\sinh z$ ——— (2)

$w = 3x^4 e^{-2y}$ ——— (3)

Differentiating these eqn's w.r.to. x, y, z, we get

$\frac{\partial u}{\partial x} = 2x e^{-y}\cosh z$, $\frac{\partial u}{\partial y} = -x^2 e^{-y}\cosh z$, $\frac{\partial u}{\partial z} = x^2 e^{-y}\sinh z$

$\frac{\partial v}{\partial x} = 2x e^{-y}\sinh z$ $\frac{\partial v}{\partial y} = -x^2 e^{-y}\sinh z$ $\frac{\partial v}{\partial z} = x^2 e^{-y}\cosh z$

$\frac{\partial w}{\partial x} = 12x^3 e^{-2y}$ $\frac{\partial w}{\partial y} = -6x^4 e^{-2y}$ $\frac{\partial w}{\partial z} = 0$

$$\therefore \frac{\partial(u,v,w)}{\partial(x,y,z)} = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} & \frac{\partial u}{\partial z} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} & \frac{\partial v}{\partial z} \\ \frac{\partial w}{\partial x} & \frac{\partial w}{\partial y} & \frac{\partial w}{\partial z} \end{vmatrix} = \begin{vmatrix} 2x e^{-y}\cosh z & -x^2 e^{-y}\cosh z & x^2 e^{-y}\sinh z \\ 2x e^{-y}\sinh z & -x^2 e^{-y}\sinh z & x^2 e^{-y}\cosh z \\ 12x^3 e^{-2y} & -6x^4 e^{-2y} & 0 \end{vmatrix}$$

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$$= (x e^{-y}) \cdot (x e^{-y}) \cdot (x^3 e^{-2y}) \begin{vmatrix} 2\cosh z & -x\cosh z & x\sinh z \\ 2\sinh z & -x\sinh z & x\cosh z \\ 2 & -x & 0 \end{vmatrix}$$

$$= 6x^5 e^{-4y} \left\{ (2)(x)(x) \begin{vmatrix} \cosh z & \cosh z & \sinh z \\ \sinh z & \sinh z & \cosh z \\ 1 & 1 & 0 \end{vmatrix} \right.$$

$$= 12(-x^7) e^{-4y} \left(\cosh z [-\cosh z] - \cosh z(-\cosh z) + \sinh z(-\sinh z + \sinh z) \right)$$

$$= -12x^7 e^{-4y} (-2) \left(\cancel{\cosh^2 z + \sinh^2 z} \right) \left[-\cosh^2 z + \cosh^2 z - \sinh^2 z + \sinh^2 z \right)$$

$$= -24 x^7 e^{-4y} (0)$$

$$\frac{\partial(u,v,w)}{\partial(x,y,z)} = 0$$

This shows that u, v, w are functionally related. (dependent)

Relation betn u, v, w

since $u = x^2 e^{-y} \cosh z$

$\Rightarrow u^2 = x^4 e^{-2y} \cosh^2 z$ ——— (4)

and $v = x^2 e^{-y} \sinh z$

$\Rightarrow v^2 = x^4 e^{-2y} \sinh^2 z$ ——— (5)

Now adding (4) and (5) we get

$$u^2 - v^2 = x^4 e^{-2y} \cosh^2 z - x^4 e^{-2y} \sinh^2 z$$

$$= x^4 e^{-2y} (\cosh^2 z - \sinh^2 z) = x^2 e^{-2y} \cdot = \frac{w}{3}$$

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$\Rightarrow u^2 - v^2 = \frac{w}{3}$ (i from (3))

$\Rightarrow \boxed{3u^2 - 3v^2 = w}$

This is the required relation betn $u, v,$ and w

Ex: If $u = 3x + 2y - z$, $v = x - 2y + z$

$w = x(x + 2y - z)$, show that they are functionally related and find the relation.

Soln: Let

$$u = 3x + 2y - z \quad \text{---(1)}$$

$$v = x - 2y + z \quad \text{---(2)}$$

$$w = x(x + 2y - z) \quad \text{---(3)}$$

$$= x^2 + 2xy - xz$$

Differentiating these equations partially w.r.to. x, y, z

$$\frac{\partial u}{\partial x} = 3 \qquad \frac{\partial u}{\partial y} = 2 \qquad \frac{\partial u}{\partial z} = -1$$

$$\frac{\partial v}{\partial x} = 1 \qquad \frac{\partial v}{\partial y} = -2 \qquad \frac{\partial v}{\partial z} = 1$$

$$\frac{\partial w}{\partial x} = 2x + 2y - z, \quad \frac{\partial w}{\partial y} = 2x, \quad \frac{\partial w}{\partial z} = -x$$

$$\therefore \frac{\partial(u,v,w)}{\partial(x,y,z)} = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} & \frac{\partial u}{\partial z} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} & \frac{\partial v}{\partial z} \\ \frac{\partial w}{\partial x} & \frac{\partial w}{\partial y} & \frac{\partial w}{\partial z} \end{vmatrix} = \begin{vmatrix} 3 & 2 & -1 \\ 1 & -2 & 1 \\ 2x+2y-z & 2x & -x \end{vmatrix}$$

$$= 3(2x - 2x) - 2(-x - 2x - 2y + z) - 1(2x + 4x + 4y - z)$$

$$= 0 - 2x + 4x + 4y - z - 2x + 4x + 4y + z$$

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$\Rightarrow \frac{\partial(u,v,w)}{\partial(x,y,z)} = -4x \neq 0$

This shows that u, v, w are not functionally related.

If $u = \frac{yz}{x}$, $v = \frac{xz}{y}$ $w = \frac{xy}{z}$

find $\frac{\partial(x,y,z)}{\partial(u,v,w)}$. Are u, v, w are functionally related? if so. find relationship.

Sol: let $u = \frac{yz}{x}$; $v = \frac{xz}{y}$; $w = \frac{xy}{z}$

$\Rightarrow \frac{\partial u}{\partial x} = -\frac{yz}{x^2}$, $\frac{\partial u}{\partial y} = \frac{z}{x}$, $\frac{\partial u}{\partial z} = \frac{y}{x}$

$\frac{\partial v}{\partial x} = \frac{z}{y}$, $\frac{\partial v}{\partial y} = -\frac{xz}{y^2}$, $\frac{\partial v}{\partial z} = \frac{x}{y}$

$\frac{\partial w}{\partial x} = \frac{y}{z}$, $\frac{\partial w}{\partial y} = \frac{x}{z}$, $\frac{\partial w}{\partial z} = -\frac{xy}{z^2}$

$$\therefore \frac{\partial(u,v,w)}{\partial(x,y,z)} = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} & \frac{\partial u}{\partial z} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} & \frac{\partial v}{\partial z} \\ \frac{\partial w}{\partial x} & \frac{\partial w}{\partial y} & \frac{\partial w}{\partial z} \end{vmatrix} = \begin{vmatrix} -\frac{yz}{x^2} & \frac{z}{x} & \frac{y}{x} \\ z/y & -\frac{xz}{y^2} & \frac{x}{y} \\ y/z & x/z & -\frac{xy}{z^2} \end{vmatrix}$$

$$= \frac{1}{x}\frac{1}{y}\frac{1}{z} \begin{vmatrix} -\frac{yz}{x} & z & y \\ z & -\frac{xz}{y} & x \\ y & x & -\frac{xy}{z} \end{vmatrix}$$

$$= \frac{1}{(xyz)} \cdot \left[\left(\frac{-yz}{x}\right)\left(x^2 - x^2\right) - z(-xy - xy) + y(zx + xz) \right]$$

$$= \frac{1}{xyz} \cdot \left[0 + 2xyz + 2(xyz)\right] = \frac{4(xyz)}{(xyz)} = 4$$

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Since $\frac{\partial(u,v,w)}{\partial(x,y,z)} = 4 \neq 0$

$\therefore \Rightarrow$ u, v, w are not functionally related.

Now. To find $\frac{\partial(x,y,z)}{\partial(u,v,w)}$

We know that

$$\frac{\partial(u,v,w)}{\partial(x,y,z)} \times \frac{\partial(x,y,z)}{\partial(u,v,w)} = 1$$

$$\Rightarrow \quad 4 \cdot \frac{\partial(x,y,z)}{\partial(u,v,w)} = 1$$

$$\Rightarrow \quad \boxed{\frac{\partial(x,y,z)}{\partial(u,v,w)} = \frac{1}{4}}$$

Ex: Given $u = 4x^2 + 9y^2 + 16z^2$, $v = 12xy + 16xz + 24yz$

$w = 2x + 3y + 4z$

Are u, v and w functionally related. If so find the rel

Sol^n: $\Rightarrow$ let

$$u = 4x^2 + 9y^2 + 16z^2 \quad \text{---} (1)$$

$$v = 12xy + 16xz + 24yz \quad \text{---} (2)$$

$$w = 2x + 3y + 4z \quad \text{---} (3)$$

$\Rightarrow \frac{\partial u}{\partial x} = 8x$, $\frac{\partial u}{\partial y} = 18y$, $\frac{\partial u}{\partial z} = 32z$

$\frac{\partial v}{\partial x} = 12y + 16z$, $\frac{\partial v}{\partial y} = 12x + 24z$, $\frac{\partial v}{\partial z} = 16x + 24y$

$\frac{\partial w}{\partial x} = 2$, $\frac{\partial w}{\partial y} = 3$, $\frac{\partial w}{\partial z} = 4$

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$$\frac{\partial(u,v,w)}{\partial(x,y,z)} = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} & \frac{\partial u}{\partial z} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} & \frac{\partial v}{\partial z} \\ \frac{\partial w}{\partial x} & \frac{\partial w}{\partial y} & \frac{\partial w}{\partial z} \end{vmatrix}$$

$$= \begin{vmatrix} 8x & 18y & 32z \\ 4(3y+4z) & 4(3x+6z) & 4(4x+6y) \\ 2 & 3 & 4 \end{vmatrix}$$

$$\frac{\partial(u,v,w)}{\partial(x,y,z)} = 0$$

∴ This shows that u, v, w are functionally related.

Now, Relation bet^n u, v, w

Let $w = 2x + 3y + 4z$

on squaring we get

$$w^2 = (2x+3y+4z)^2 = [(2x+3y)+(4z)]^2$$

$$= (2x+3y)^2 + 2(2x+3y)(4z) + (4z)^2$$

$$= 4x^2 + 12xy + 9y^2 + 16xz + 24yz + 16z^2$$

$$= \underline{4x^2 + 9y^2 + 16z^2} + \underline{12xy + 16xz + 24yz}$$

$$= u + v$$

$$\Rightarrow \boxed{u + v = w^2}$$ is the required relation.

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(HW)

Ex: Are the followings functionally related ~~functions~~ ? Justify your Answer If so, find ~~their~~ their relation.

(1) $u = \frac{x+y}{1-xy}$, $v = \tan^{-1}x + \tan^{-1}y$

∴ Relation $\boxed{u = \tan v}$

(2) $u = x+y+z$, $v = x^2+y^2+z^2-2xy-2yz-2zx$
$w = x^3+y^3+z^3-3xyz$

Ans $\boxed{4w = 3uv + u^3}$

(3) ~~$u = \frac{x-y}{x+z}$~~ , ~~$v = \frac{x+z}{y+z}$~~

Ans $\boxed{u + \frac{1}{v} = 1}$ or $\boxed{v = uv + 1}$

Hint: consider $w = z$ and find $\frac{\partial(u,v,w)}{\partial(x,y,z)}$

(4) $u = \frac{3x^2}{2(y+z)}$, $v = \frac{2(y+z)}{3(x-y)^2}$, $w = \frac{x-y}{x}$

Ans $\boxed{uvw^2 = 1}$

(5) $u = \sin x + \sin y$, $v = \sin(x+y)$

$\boxed{\text{not Related. i.e } \frac{\partial(u,v)}{\partial(x,y)} \neq 0}$

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Taylor series of function of Two variable

let $f(x,y)$ be the function of x & y.
and h & k are constants.

Then

$$f(x+h, y+k) = f(x,y) + \left(h\frac{\partial f}{\partial x} + k\frac{\partial f}{\partial y}\right) + \frac{1}{2!}\left[h^2\frac{\partial^2 f}{\partial x^2} + 2hk\frac{\partial^2 f}{\partial x \partial y} + k^2\frac{\partial^2 f}{\partial y^2}\right] + \frac{1}{3!}\left[h^3\frac{\partial^3 f}{\partial x^3} + 3h^2k\frac{\partial^3 f}{\partial x^2 \cdot \partial y} + 3hk^2\frac{\partial^3 f}{\partial x \partial y^2} + k^3\frac{\partial^3 f}{\partial y^3}\right] + \dots$$

This is the taylor series expantion of $f(x+h, y+k)$ in ascending power of h & k.

Case (I) If $x = a$, $y = b$ and $h = x-a$, & $k = y-b$ Then

$$f(x,y) = f(a,b) + \left[(x-a)\left(\frac{\partial f}{\partial x}\right)_{(a,b)} + (y-b)\left(\frac{\partial f}{\partial y}\right)_{(a,b)}\right] + \frac{1}{2!}\left[(x-a)^2\left(\frac{\partial^2 f}{\partial x^2}\right)_{(a,b)} + 2(x-a)(y-b)\left(\frac{\partial^2 f}{\partial x \partial y}\right)_{(a,b)} + (y-b)^2\left(\frac{\partial^2 f}{\partial y^2}\right)_{(a,b)}\right] + \dots$$

This is called Taylor Series expantion of $f(x,y)$ in power of $(x-a)$ & $(y-b)$ or about pt (a,b)

Case (II): put $a=0, b=0$

$$f(x,y) = f(0,0) + \left[x\left(\frac{\partial f}{\partial x}\right)_{(0,0)} + y\left(\frac{\partial f}{\partial y}\right)_{(0,0)}\right] + \frac{1}{2!}\left[x^2\left(\frac{\partial^2 f}{\partial x^2}\right)_{(0,0)} + 2xy\left(\frac{\partial^2 f}{\partial x \partial y}\right)_{(0,0)} + y^2\left(\frac{\partial^2 f}{\partial y^2}\right)_{(0,0)}\right] + \dots$$

This is known as maclaurins series expantion of $f(x,y)$ in power of x & y. (about origin $(0,0)$).

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Ex: Expand $x^2y + 3y - 2$ in power of $(x-1)$ and $(y+2)$ as for as term of second degree

Soln: let $f(x,y) = x^2y + 3y - 2$

put $a=1,\ b=-2$ $\therefore f(a,b) = f(1,-2) = (1)^2(-2) + 3(-2) - 2$
$= -10$

Now $\frac{\partial f}{\partial x} = 2xy \Rightarrow \left(\frac{\partial f}{\partial x}\right)_{(1,-2)} = -4$

$\frac{\partial f}{\partial y} = x^2 + 3 \Rightarrow \left(\frac{\partial f}{\partial y}\right)_{(1,-2)} = 4$

$\frac{\partial^2 f}{\partial x^2} = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial x}\right) = 2y \Rightarrow \left(\frac{\partial^2 f}{\partial x^2}\right)_{(1,-2)} = -4$

$\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial y}\right) = \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial x}\right) = 2x \Rightarrow \left(\frac{\partial^2 f}{\partial x \partial y}\right)_{(1,-2)} = 2$

$\frac{\partial^2 f}{\partial y^2} = \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial y}\right) = 0 \Rightarrow \left(\frac{\partial^2 f}{\partial y^2}\right)_{(1,-2)} = 0$

and so on.

Now by Taylor's expansion of $f(x,y)$ in power of $(x-a)$ and $(y-b)$ we have

$$f(x,y) = f(a,b) + \left[(x-a)\left(\frac{\partial f}{\partial x}\right)_{(a,b)} + (y-b)\left(\frac{\partial f}{\partial y}\right)_{(a,b)}\right] + \frac{1}{2!}\left[(x-a)^2\left(\frac{\partial^2 f}{\partial x^2}\right)_{(a,b)} + 2(x-a)(y-b)\left(\frac{\partial^2 f}{\partial x \partial y}\right)_{(a,b)} + (y-b)^2\left(\frac{\partial^2 f}{\partial y^2}\right)_{(a,b)}\right] + \cdots$$

or $a=1, b=-2$

$= -10 + \left[-4(x-1) + 4(y+2)\right] + \frac{1}{2!}\left[-4(x-1)^2 + 4(x-1)(y+2) + 0\right] + \cdots$

$= -10 - 4(x-1) + 4(y+2) - 2(x-1)^2 + 2(x-1)(y+2) + \cdots$

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Ex: obtain Taylor's expansion of $\tan^{-1} y/x$ about $(1,1)$ upto third degree term.

Soln let $f(x,y) = \tan^{-1}(y/x) \Rightarrow f(1,1) = \tan^{-1}(1) = \tan^{-1}(1) = \pi/4$

$$\Rightarrow \frac{\partial f}{\partial x} = \frac{1}{1+(y/x)^2}\cdot\left(\frac{-y}{x^2}\right) = \frac{-y}{(x^2+y^2)} \Rightarrow \left(\frac{\partial f}{\partial x}\right)_{(1,1)} = \frac{-1}{2}$$

$$\Rightarrow \frac{\partial f}{\partial y} = \frac{1}{1+(y/x)^2}\cdot\left(\frac{1}{x}\right) = \frac{x}{(x^2+y^2)} \Rightarrow \left(\frac{\partial f}{\partial y}\right)_{(1,1)} = 1/2$$

$$\Rightarrow \frac{\partial^2 f}{\partial x^2} = \frac{2xy}{(x^2+y^2)^2} \Rightarrow \left(\frac{\partial^2 f}{\partial x^2}\right)_{(1,1)} = \frac{2}{4} = \frac{1}{2}$$

$$\Rightarrow \frac{\partial^2 f}{\partial x\,\partial y} = \frac{y^2-x^2}{(x^2+y^2)^2} \Rightarrow \left(\frac{\partial^2 f}{\partial x\,\partial y}\right)_{(1,1)} = 0$$

$$\Rightarrow \frac{\partial^2 f}{\partial y^2} = \frac{-2xy}{(x^2+y^2)} \Rightarrow \left(\frac{\partial^2 f}{\partial y^2}\right)_{(1,1)} = -1/2$$

$$\Rightarrow \frac{\partial^3 f}{\partial x^3} = 2y\left[\frac{(x^2+y^2)^2 * - x\,2(x^2+y^2).2x}{(x^2+y^2)^4}\right]$$

$$= 2y\left[\frac{1}{(x^2+y^2)^2} - \frac{4x^2}{(x^2+y^2)^3}\right] \Rightarrow \left(\frac{\partial^3 f}{\partial x^3}\right)_{(1,1)} = 2\left(\frac{1}{4}-\frac{4}{8}\right) = -1/2$$

$$\Rightarrow \frac{\partial^3 f}{\partial x^2\,\partial y} = \frac{\partial}{\partial y}\left(\frac{\partial^2 f}{\partial x^2}\right) = 2x\left[\frac{(x^2+y^2)^2 - y\,2(x^2+y^2)+2y}{(x^2+y^2)^4}\right]$$

$$= \Rightarrow \left(\frac{\partial^3 f}{\partial x^2\,\partial y}\right)_{(1,1)} = -1/2$$

$$\Rightarrow \frac{\partial^3 f}{\partial x\,\partial y^2} = \frac{\partial}{\partial x}\left(\frac{\partial^2 f}{\partial y^2}\right) =$$

$$= -2y\left[\frac{(x^2+y^2) - x\,2(x^2+y^2)\,2x}{(x^2+y^2)^4}\right]$$

$$\Rightarrow \left(\frac{\partial^3 f}{\partial x\,\partial y^2}\right)_{(1,1)} = 1/2.$$

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$$\frac{\partial^3 f}{\partial y^3} = -2x\left[\frac{(x^2+y^2)^2 - 2y(x^2+y^2)(2x)}{(x^2+y^2)^4}\right]$$

$$= 1/2$$

and so on

Now we know that

Expansion of $f(x,y)$ in powers of $(x-a), (y-b)$ is given by Taylor's thm as follows

$$f(x,y) = f(a,b) + \left[(x-a)\left(\frac{\partial f}{\partial x}\right)_{(a,b)} + (y-b)\left(\frac{\partial f}{\partial y}\right)_{(a,b)}\right]$$

$$+ \frac{1}{2!}\left[(x-a)^2\left(\frac{\partial^2 f}{\partial x^2}\right)_{(a,b)} + 2(x-a)(y-b)\left(\frac{\partial^2 f}{\partial x \partial y}\right)_{(a,b)} + (y-b)^2\left(\frac{\partial^2 f}{\partial y^2}\right)\right]$$

$$+ \frac{1}{3!}\left[(x-a)^3\left(\frac{\partial^3 f}{\partial x^3}\right)_{(a,b)} + 3(x-a)^2(y-b)\left(\frac{\partial^3 f}{\partial x^2 \partial y}\right)_{(a,b)} + 3(x-a)(y-b)^2\left(\frac{\partial^3 f}{\partial x \partial y^2}\right)\right.$$

$$\left. + (y-b)^3\left(\frac{\partial^3 f}{\partial y^3}\right)_{(a,b)}\right] + \cdots$$

Now about $(a,b) = (1,1)$.

$$f(x,y) = \tan^{-1} y/x = \pi/4 + \left[(x-1)(-1/2) + (y-1)(1/2)\right]$$

$$+ \frac{1}{2!}\left[(x-1)^2\left(\frac{1}{2}\right) + 0 + (y-1)^2\left(\frac{-1}{2}\right)\right]$$

$$+ \frac{1}{3!}\left[(x-1)^3\left(\frac{-1}{2}\right) + 3(x-1)^2(y-1)\left(\frac{-1}{2}\right) + 3(x-1)(y-1)^2\left(\frac{1}{2}\right)\right.$$

$$\left. + (y-1)^3\left(\frac{1}{2}\right)\right]$$

$$\tan^{-1}(y/x) = \pi/4 + \left(\frac{(y-x)}{2}\right) + \frac{1}{4}\left(\frac{x^2 + 2(y-x) + 2}{2}\right) + \cdots$$

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Ex: Expand $xy^2 + \sin xy$ at the point $(1, \pi/2)$ up to second degree term.

Soln: $\Rightarrow$ let $f(x,y) = xy^2 + \sin(xy)$

$$\Rightarrow f(1, \pi/2) = \pi^2/4 + \sin(\pi/2) = (\pi^2/4 + 1)$$

$$\Rightarrow \frac{\partial f}{\partial x} = y^2 + \cos(xy)\cdot y \Rightarrow \left(\frac{\partial f}{\partial x}\right)_{(1,\pi/2)} = \frac{\pi^2}{4} + \frac{\pi}{2}\cdot\cos\left(\frac{\pi}{2}\right) = \frac{\pi^2}{4}$$

$$\frac{\partial f}{\partial y} = 2xy + x\cos(xy) \Rightarrow \left(\frac{\partial f}{\partial y}\right)_{(1,\pi/2)} = \pi + 0 = \pi$$

$$\frac{\partial^2 f}{\partial x^2} = -y^2\sin(xy) \Rightarrow \left(\frac{\partial^2 f}{\partial x^2}\right)_{(1,\pi/2)} = -\frac{\pi^2}{4}$$

$$\frac{\partial^2 f}{\partial x \partial y} = 2y + [y(-\sin(xy)\cdot x) + \cos(xy)]$$

$$\Rightarrow \left(\frac{\partial^2 f}{\partial x \partial y}\right)_{(1,\pi/2)} = \pi\left(-\frac{\pi}{2}(1) + 0\right] = -\frac{\pi^2}{2} = -\pi/2$$

$$\frac{\partial^2 f}{\partial y^2} = 2x + -x^2\cdot\sin(xy) = \pi/2$$

$$\Rightarrow \left(\frac{\partial^2 f}{\partial y^2}\right)_{(1,\pi/2)} = 2 - 1(1) = 1$$

and so on.

Now, by Taylor's thm. expansion of $f(x,y)$ in powers $(x-a)$ & $(y-b)$ is given by

$$f(x,y) = f(a,b) + \left[(x-a)\left(\frac{\partial f}{\partial x}\right)_{(a,b)} + (y-b)\left(\frac{\partial f}{\partial y}\right)_{(a,b)}\right]$$

$$+ \frac{1}{2!}\left[(x-a)^2\left(\frac{\partial^2 f}{\partial x^2}\right)_{(a,b)} + 2(x-a)(y-b)\left(\frac{\partial^2 f}{\partial x \partial y}\right)_{(a,b)} + (y-b)^2\left(\frac{\partial^2 f}{\partial y^2}\right)_{(a,b)}\right] + \cdots$$

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$\therefore$ at pt $(1, \pi/2) = (a, b)$

$f(x,y) = xy^2 + \sin xy$

$= \left(1 + \frac{\pi^2}{4}\right) + \left[(x-1)\frac{\pi^2}{4} + (y - \pi/2)\pi\right]$

$+ \frac{1}{2!}\left[(x-1)^2\left(\frac{-\pi^2}{4}\right) + 2(x-1)(y-\pi/2)\cdot\pi/2 + (y-\frac{\pi}{2})^2 \cdot 1\right] + \cdots$

$y^2 + \sin xy = 1 + \frac{\pi^2}{4} + \frac{\pi^2}{4}(x-1) + \pi(y-\pi/2) - \frac{\pi^2}{8}(x-1)^2 + \frac{\pi}{2}(x-1)(y-\pi/2)$

$+ \frac{1}{2}(y-\pi/2)^2 + \cdots$

Ex: Expand $e^x \sin y$ in power of x and y as far as term of third degree.

Soln: let $f(x,y) = e^x \sin y \Rightarrow f(0,0) = e^0 \sin 0 = 0$

$\frac{\partial f}{\partial x} = e^x \sin y \Rightarrow \left(\frac{\partial f}{\partial x}\right)_{(0,0)} = 0$

$\frac{\partial f}{\partial y} = e^x \cos y \Rightarrow \left(\frac{\partial f}{\partial y}\right)_{(0,0)} = e^0 \cos 0 = 1 \cdot 1 =$

$\frac{\partial^2 f}{\partial x^2} = e^x \sin y \Rightarrow \left(\frac{\partial^2 f}{\partial x^2}\right)_{(0,0)} = 0$

$\frac{\partial^2 f}{\partial x \partial y} = e^x \cos y \Rightarrow \left(\frac{\partial^2 f}{\partial x \partial y}\right)_{(0,0)} = 1$

$\frac{\partial^2 f}{\partial y \partial x} = -e^x \sin y \Rightarrow \left(\frac{\partial^2 f}{\partial y^2}\right)_{(0,0)} = 0$

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$\frac{\partial^3 f}{\partial x^3} = e^x \sin y \Rightarrow \left(\frac{\partial^3 f}{\partial x^3}\right)_{(0,0)} = 0$

$\frac{\partial^3 f}{\partial x^2 \partial y} = e^x \cos y \Rightarrow \left(\frac{\partial^3 f}{\partial x^2 \partial y}\right)_{(0,0)} = 1$

$\frac{\partial^3 f}{\partial x \partial y^2} = -e^x \sin y \Rightarrow \left(\frac{\partial^3 f}{\partial x \partial y}\right)_{(0,0)} = 0$

$\frac{\partial^3 f}{\partial y^3} = -e^x \cos y \Rightarrow \left(\frac{\partial^3 f}{\partial y^3}\right)_{(0,0)} = -1$

and so on.

Now by Maclaurin's thm we have

$$f(x,y) = f(0,0) + \left[x\left(\frac{\partial f}{\partial x}\right)_{(0,0)} + y\left(\frac{\partial f}{\partial y}\right)_{(0,0)}\right] + \frac{1}{2!}\left[x^2\left(\frac{\partial^2 f}{\partial x^2}\right)_{(0,0)} + 2xy\left(\frac{\partial^2 f}{\partial x \partial y}\right)_{(0,0)} + y^2\left(\frac{\partial^2 f}{\partial y^2}\right)_{(0,0)}\right] + \frac{1}{3!}\left[x^3\left(\frac{\partial^3 f}{\partial x^3}\right)_{(0,0)} + 3x^2y\left(\frac{\partial^3 f}{\partial x^2 \partial y}\right)_{(0,0)} + 3xy^2\left(\frac{\partial^3 f}{\partial x \partial y^2}\right)_{(0,0)} + y^3\left(\frac{\partial^3 f}{\partial y^3}\right)_{(0,0)}\right] + \cdots$$

$$\Rightarrow e^x \sin y = 0 + [x(0) + y(1)] + \frac{1}{2!}[0 + 2xy(1) + 0] + \frac{1}{3!}[0 + 3x^2y(1) + 0 + y^3(-1)] + \cdots$$

$$= y + xy + \frac{1}{3!}(3x^2y) - \frac{1}{3!}y^3 + \cdots$$

$$= y + xy + \frac{1}{2}x^2y + \frac{y^3}{6} + \cdots$$

$$\boxed{e^x \sin y = y + xy + \frac{x^2y}{2} - \frac{y^3}{6} + \cdots}$$

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H.W.:

① Expand $\sin(xy)$ in power of $(x-1)$ and $(y-\pi/2)$ upto and for as second degree term.

② expand $e^x \log(1+y)$ upto six term

③ expand y^x in the neighbourhood of $(1,1)$ upto term of second degree.

④ expand $\sin(x+h)\cdot(y+k)$ by Taylor's thm.

⑤ Expand $x^2y + 2y - x^2 - 2$ in power of $(x-1)$ and $(y+1)$ by Taylor's thm.

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Maxima and Minima of Function of two variables

Let $u = f(x,y)$ be a continuous function of x and y.

Then u will be maximum at $x=a, y=b$ if $f(a,b) > f(a+h, b+k)$ and will be minimum at $x=a, y=b$ if $f(a,b) < f(a+h, b+k)$ for small positive or negative values of h and k.

* The point at which function $f(x,y)$ is either maximum or minimum is known as stationary point.
* The value of the function at stationary point is known as extreme (maximum or minimum) value of the function $f(x,y)$.

Working rule: To determine the maxima and minima (extreme value) of a function $f(x,y)$

Step (I): ⇒ Solve $\frac{\partial f}{\partial x} = 0$ & $\frac{\partial f}{\partial y} = 0$ simultaneously for x and y.

Consider $x=a$ & $y=b$ ($f(a,b)$ is called stationary point)

Step (II): ⇒ Obtain the values of $r = \frac{\partial^2 f}{\partial x^2}$, $s = \frac{\partial^2 f}{\partial x \partial y}$, $t = \frac{\partial^2 f}{\partial y^2}$ at stationary points

Step (III): ⇒ (i) If $rt - s^2 > 0$ and $r < 0$ (or $t < 0$), then $f(x,y)$ is maximum at (a,b). and maximum value of the function $f(x,y)$ is $f(a,b)$.

(ii) If $rt - s^2 > 0$ and $r > 0$ or $(t > 0)$ then $f(x,y)$ is minimum at (a,b) and minimum value of the function $f(x,y)$ is $f(a,b)$.

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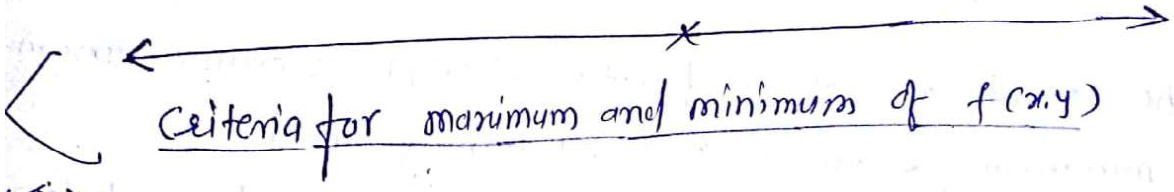
(iii) If $rt - s^2 < 0$ at (a,b). then $f(x,y)$ is neither maximum nor minimum. at (a,b) such point is known as <u>saddle point</u>

(iv) If $rt - s^2 = 0$ at (a,b) then no conclusion can be made about the extreme values of $f(x,y)$ and further investigation is required.



Criteria for maximum and minimum of $f(x,y)$

Thm: The necessary condn for $f(a,b)$ to be an extremum value of $f(x,y)$ are

$$\left(\frac{\partial f}{\partial x}\right)_{(a,b)} = 0, \quad \left(\frac{\partial f}{\partial y}\right)_{(a,b)} = 0$$

Also if $r = \left(\frac{\partial^2 f}{\partial x^2}\right)$, $s = \left(\frac{\partial^2 f}{\partial x \partial y}\right)_{(a,b)}$, $t = \left(\frac{\partial^2 f}{\partial y^2}\right)_{(a,b)}$

Then

(i) $f(a,b)$ is maximum if $(rt - s^2) > 0$ and $r < 0$

(ii) $f(a,b)$ is minimum if $(rt - s^2) > 0$ and $r > 0$

(iii) $f(a,b)$ is not an extreme value if $(rt - s^2) < 0$ (in such case pt. (a,b) is called saddle pt.)

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Ex :⇒ Show that the function $f(x,y) = x^3+y^3 - 63(x+y) + 12xy$ is maximum at $(-7,-7)$ and minimum at $(3,3)$

Soln :⇒ Let $f(x,y) = x^3+y^3 - 63(x+y) + 12xy$

$$\Rightarrow \frac{\partial f}{\partial x} = 3x^2 - 63 + 12y$$

$$\frac{\partial f}{\partial y} = 3y^2 - 63 + 12x$$

and $r = \frac{\partial^2 f}{\partial x^2} = 6x$

$$s = \frac{\partial^2 f}{\partial x \partial y} = 12$$

$$t = \frac{\partial^2 f}{\partial y^2} = 6y$$

Now, Extreme value is given by,

$$\frac{\partial f}{\partial x} = 0, \quad \frac{\partial f}{\partial y} = 0$$

i.e $3x^2 + 12y - 63 = 0$ ——— (1)

$3y^2 + 12x - 63 = 0$ ——— (2)

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$\Rightarrow$ or

$$x^2 + 4y - 21 = 0 \quad \text{---(1)}$$
$$y^2 + 4x - 21 = 0 \quad \text{---(2)}$$

by subtracting eqn (1) and (2) we get.

$$(x^2 - y^2) + 4(y - x) = 0$$

$$\Rightarrow (x^2 - y)(x + y) + 4(y - x) = 0$$

$$\Rightarrow (x - y)[(x + y) - 4] = 0$$

$$\Rightarrow (x - y) = 0 \quad \text{and} \quad (x + y) - 4 = 0$$

$$\Rightarrow \boxed{x = y} \quad \text{or} \quad \boxed{x + y = 4}$$

$x^3 + 4xy = 21x$
$y^3 + 4xy = 21y$
$(x^3 - y^3) = 21(x - y)$
$(x^2 + xy + y^2) = 21$

for $x = y$

$$\text{(1)} \Rightarrow x^2 + 4x - 21 = 0$$
$$\Rightarrow (x + 7)(x - 3) = 0$$
$$\Rightarrow \boxed{x = -7}, \ \boxed{x = 3}$$

$\therefore \boxed{y = -7}, \ \boxed{y = 3}$

Now for $x + y = 4 \Rightarrow x = 4 - y$

$x^2 + 4(4 - x) - 21 =$
$\Rightarrow x^2 - 4x + 5 = 0$

$$\text{(2)} \Rightarrow y^2 + 4(4 - y) - 21 = 0$$
$$\Rightarrow y^2 - 4y - 5 = 0$$
$$\Rightarrow (y - 5)(y + 1) = 0$$

$\therefore$ i.e. $\boxed{y = 5}$ or $\boxed{y = -1}$

Hence $x = 4 - 5 = -1 \Rightarrow \boxed{x = -1}$

or $x = 4 - (-1) = 5 \quad \boxed{x = 5}$

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Hence

$(-7,-7)$, $(3,3)$, $(-1,5)$, $(5,-1)$

may be possible extreme of $f(x,y)$.

	$(-7,-7)$	$(3,3)$	$(-1,5)$	$(5,1)$
$r=6x$	-42	18	-6	30
$s=12$	12	12	12	12
$t=6y$	-42	18	30	6
$rt-s^2$	1620	180	-324	-324

Since at pt. $(-7,-7)$

$r<0$ and $rt-s^2 = 1620 > 0$

$\Rightarrow$ $f(x,y)$ is maximum at $(-7,-7)$.

and at $(3,3)$

$r>0$, and $rt-s^2 > 0$

$\Rightarrow$ $f(x,y)$ is minimum at $(3,3)$.

at $(-1,5)$: $r<0$ and $rt-s^2<0$

$\Rightarrow$ $f(x,y)$ is neither maximum nor minimum at $(-1,5)$. therefor this point is called Saddle point.

at $(1,5)$: $r>0$, $rt-s^2<0$ $\Rightarrow$ Saddle point

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Ex: Find the minimum value of $x^2+y^2+z^2$, given that

$$ax+by+cz=p$$

Sol^n :⇒ Let $f(x,y,z) = x^2+y^2+z^2$ —(1)

Here $f(x,y,z)$ can be reduced in ~~two~~ function of two variable with the help of given relation

$$ax+by+cz=p.$$

$$\Rightarrow z = \frac{p-ax-by}{c}$$

$$\therefore (1) \Rightarrow f(x,y) = x^2+y^2+\frac{(p-ax-by)^2}{c^2} \quad \text{—(2)}$$

$$\Rightarrow \frac{\partial f}{\partial x} = 2x + 2\frac{(p-ax-by)}{c^2}\cdot(-a)$$

$$\frac{\partial f}{\partial x} = 2x - \frac{2a}{c^2}(p-ax-by)$$

and $$\frac{\partial f}{\partial y} = 2y - \frac{2b}{c^2}(p-ax-by)$$

Similarly. $$r = \frac{\partial^2 f}{\partial x^2} = 2 + \frac{2a^2}{c^2}(p-ax+by) = 2\left(1+\frac{a^2}{c^2}\right)$$

$$s = \frac{\partial^2 f}{\partial x \partial y} = \frac{2ab}{c^2}$$

$$t = \frac{\partial^2 f}{\partial y^2} = 2 + \frac{2b^2}{c^2} = 2\left(1+\frac{b^2}{c^2}\right)$$

Now for maximum and minimum value of function $f(x,y)$ we must have

$$\frac{\partial f}{\partial x} = 0, \quad \text{and} \quad \frac{\partial f}{\partial y} = 0$$

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i.e. $2x - \frac{2a}{c^2}(p - ax - by) = 0$

i.e. $\Rightarrow x - \frac{a}{c^2}(p - ax - by) = 0$

$\Rightarrow (a^2 + c^2)x + aby - pa = 0$

and $2y - \frac{2b}{c^2}(p - ax - by) = 0$

i.e. $\Rightarrow y - \frac{b}{c^2}(p - ax - by) = 0$

$\Rightarrow (b^2 + c^2)y + abx - pb = 0$

$\Rightarrow$ ~~abx~~

Thus we have

$$(a^2 + c^2)x + aby - pa = 0 \quad \text{---} (3)$$

$$abx + (b^2 + c^2)y - pb = 0 \quad \text{---} (4)$$

Solving eqn (3) and (4) we get

$$\frac{x}{\begin{vmatrix} ab & -pa \\ b^2+c^2 & -pb \end{vmatrix}} = \frac{-y}{\begin{vmatrix} a^2+c^2 & -pa \\ ab & -pb \end{vmatrix}} = \frac{1}{\begin{vmatrix} a^2+c^2 & ab \\ ab & b^2+c^2 \end{vmatrix}}$$

$$\Rightarrow \frac{x}{-pab^2 + pa(b^2+c^2)} = \frac{-y}{[-pb(a^2+c^2) + pa^2b]} = \frac{1}{(a^2+c^2)(b^2+c^2) - a^2b^2}$$

$$\Rightarrow \frac{x}{-pab^2 + pab^2 + pac^2} = \frac{-y}{-pba^2 + pbc^2 + pa^2b} = \frac{1}{a^2b^2 + a^2c^2 + c^4 - a^2b^2}$$

$$\Rightarrow \frac{x}{pac^2} = \frac{-y}{-pbc^2} = \frac{1}{c^2(a^2+b^2+c^2)}$$

$$\Rightarrow \frac{x}{pac^2} = \frac{y}{pbc^2} = \frac{1}{c^2(a^2+b^2+c^2)}$$

$$\Rightarrow \frac{x}{pa} = \frac{y}{pb} = \frac{1}{a^2+b^2+c^2}$$

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$\therefore x = \frac{pq}{(a^2+b^2+c^2)}$ [from 1st & IIIrd ratio]

& $y = \frac{p.b}{(a^2+b^2+c^2)}$ from IInd and IIIrd ratio]

Thus the stationary point is $\left(\frac{pq}{a^2+b^2+c^2}, \frac{pb}{a^2+b^2+c^2}\right)$

Now, $rt-s^2 = 2\cdot\left(1+\frac{q^2}{c^2}\right)\cdot 2\left(1+\frac{b^2}{c^2}\right) - \left(\frac{2qb}{c^2}\right)^2$

$= 4\left(1+\frac{q^2}{c^2}\right)\left(1+\frac{b^2}{c^2}\right) - \frac{4q^2b^2}{c^4}$

$= 4\left[1+\frac{q^2b^2}{c^4}+\frac{q^2}{c^2}+\frac{b^2}{c^2}+\frac{q^2b^2}{c^4}\right]$

$= 4\left[1+\frac{b^2}{c^2}+\frac{q^2}{c^2}+\frac{q^2b^2}{c^4}-\frac{q^2b^2}{c^4}\right]$

$= 4\left[1+\frac{q^2}{c^2}+\frac{b^2}{c^2}\right]$

Now, since $r = 2\left(1+\frac{q^2}{c^2}\right) > 0$ $\because \left(\begin{matrix} a^2>0 \\ b^2>0 \end{matrix}\right)$

and also $rt-s^2 = 4\left[1+\frac{q^2}{c^2}+\frac{b^2}{c^2}\right] > 0$

Thus $f(x,y)$ is minimum ~~for the~~ at the value point $(x,y) = \left(\frac{pq}{(a^2+b^2+c^2)}, \frac{pb}{(a^2+b^2+c^2)}\right)$

To find minimum value of $f(x,y)$.

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minimum value of $f(x,y) = x^2+y^2+\frac{(p-ax-by)^2}{c^2}$

$$= \left(\frac{pa}{a^2+b^2+c^2}\right)^2 + \left(\frac{pb}{a^2+b^2+c^2}\right)^2 + \frac{\left[p - a\frac{pa}{(a^2+b^2+c^2)} - b\frac{pb}{(a^2+b^2+c^2)}\right]}{c^2}$$

$$= \frac{p^2a^2}{(a^2+b^2+c^2)^2} + \frac{p^2b^2}{(a^2+b^2+c^2)^2} + \frac{\left(p(a^2+b^2+c^2) - pa^2 - pb^2\right)^2}{c^2\,(a^2+b^2+c^2)^2}$$

$$= \frac{p^2a^2}{(a^2+b^2+c^2)^2} + \frac{p^2b^2}{(a^2+b^2+c^2)^2} + \frac{(pa^2+pb^2+pc^2-pa^2-pb^2)^2}{c^2\,(a^2+b^2+c^2)^2}$$

$$= \frac{1}{(a^2+b^2+c^2)^2}\left[p^2a^2+p^2b^2+\frac{(pc^2)^2}{c^2}\right]$$

$$= \frac{1}{(a^2+b^2+c^2)^2}\left[p^2a^2+p^2b^2+p^2c^2\right]$$

$$= \frac{p^2(a^2+b^2+c^2)}{(a^2+b^2+c^2)^2} = \frac{p^2}{(a^2+b^2+c^2)}$$

minimum value of $f(x,y) = \dfrac{p^2}{a^2+b^2+c^2}$

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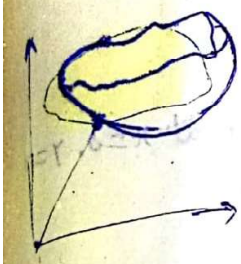
Ex':⇒ Find the point on the surface $z^2 = xy+1$ nearest to the origin. Also find that distance.

Sol":⇒ Let $P(x,y,z)$ be the any point on the surface $z^2 = xy+1$ ——— (1)

Now. The distance of a point from the origin is given by



$$D = \sqrt{(x-0)^2+(y-0)^2+(z-0)^2}$$

$$\Rightarrow D = \sqrt{x^2+y^2+z^2}$$

$$\Rightarrow D^2 = x^2+y^2+z^2$$

$$\Rightarrow f(x,y,z) = D^2 = x^2+y^2+z^2 \quad \text{——— (2)}$$

We first reduce above function as a function of two variable by using relation $z^2 = xy+1$

$$\Rightarrow f(x,y) = x^2+y^2+xy+1 \quad \text{——— (3)}$$

Now. for minimum and maximum of $f(x,y)$. necessary condn are

$$\frac{\partial f}{\partial x} = 0 \quad , \quad \frac{\partial f}{\partial y} = 0$$

i.e. $$\frac{\partial f}{\partial x} = 2x+y = 0$$

& $$\frac{\partial f}{\partial y} = 2y+x = 0$$

solving these eqn we get

$$x = 0, \quad y = 0$$

$\Rightarrow$ Thus the stationary pt is $(0,0)$

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~~at (a,b) ⇒ $(rt - s^2)$ =~~

Here, $r = \frac{\partial^2 f}{\partial x^2} = 2$

$s = \frac{\partial^2 f}{\partial x \partial y} = 1$

$t = \frac{\partial^2 f}{\partial y^2} = 2$

Now. at (0,0), $rt - s^2 = (2(2) - 1) = 3 > 0$

and $r = 2 > 0$

Thus $f(x,y)$ is minimum at (0,0). i.e at ~~x=0, y=~~

~~Since $z^2 = xy + 1$~~

putting $x=0$, $y=0$ in eqn (1). we get

$z^2 = 1 \Rightarrow \underline{z = \pm 1}$

Thus, The nearest point on the surface $z^2 = xy$ from the origin is $(0, 0, \pm 1)$

---

minimum distance

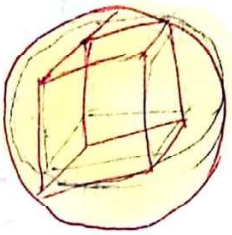
$D = \sqrt{x^2 + y^2 + z^2}$ at $(0, 0, \pm 1)$

$\Rightarrow$

$D = \sqrt{0 + 0 + (\pm 1)^2} = \sqrt{1} = 1$

minimum distance = 1 unit

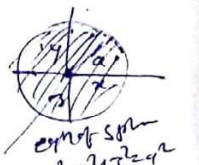
Ex :⇒ show that the rectangular solid of maximum volume that can be inscribed ing a given sphere is a cube.



Soln :⇒ Let x, y, z be the length, ~~breadth~~ breadth and height of the rectangular solid

and V be the volume

$$\therefore V = x.y.z \quad \text{——} (1)$$



Let given sphere is

$$x^2+y^2+z^2 = a^2$$

$$\Rightarrow z^2 = a^2 - x^2 - y^2$$

Substituting these value in eqn (1), we get

$$V = x.y.\left(\sqrt{a^2-x^2-y^2}\right)$$

$$\Rightarrow V^2 = x^2 y^2 (a^2-x^2-y^2)$$

Let, $f(x,y) = x^2 y^2 (a^2-x^2-y^2)$ —— (2)

Now, necessary condn for maximum of $f(x,y)$ is

$$\frac{\partial f}{\partial x} = 0, \quad \frac{\partial f}{\partial y} = 0$$

$$\Rightarrow \frac{\partial f}{\partial x} = y^2 [2x(a^2-x^2-y^2) + x^2.(-2x)] = 0$$

$$\Rightarrow 2xy^2 [a^2-x^2-y^2 - 2x^2] = 0$$

$$\Rightarrow 2xy^2 [a^2 - x^2 - y^2 - x^2] = 0$$

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$\Rightarrow 2xy^2(a^2-2x^2-y^2) = 0$

$\Rightarrow a^2-2x^2-y^2 = 0$ $\quad \because x \neq 0, y \neq 0$

$\Rightarrow 2x^2+y^2 = a^2$ ——— (3)

Because x and y are sides of rectangular solid. So it can not be zero.

and $\frac{\partial f}{\partial y} = x^2[2y(a^2-x^2-y^2) + y^2(-2y)] = 0$

$\Rightarrow 2x^2y^2(a^2-x^2-2y^2) = 0$

$\Rightarrow a^2-x^2-2y^2 = 0$ $\quad (\because x \neq 0, y \neq 0)$

$\Rightarrow x^2+2y^2 = a^2$ ——— (4)

Solving eqn (3) and (4) we get

$x = \frac{a}{\sqrt{3}}, \quad y = \frac{a}{\sqrt{3}}$

and $\therefore z = \sqrt{a^2-x^2-y^2} = \sqrt{a^2-\frac{a^2}{3}-\frac{b^2}{3}} = \frac{a}{\sqrt{3}}$

Thus the stationary point are

$\left(\frac{a}{\sqrt{3}}, \frac{a}{\sqrt{3}}, \frac{a}{\sqrt{3}}\right)$

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Now $r = \frac{\partial^2 f}{\partial x^2} = 2q^2y^2 - 12x^2y^2 - 2y^4$

$s = \frac{\partial^2 f}{\partial x \partial y} = 4q^2xy - 8x^3y - 8xy^3$

$t = \frac{\partial^2 f}{\partial y^2} = 2q^2x^2 - 2x^4 - 12x^2y^2$

$\therefore$ at $\left(\frac{q}{\sqrt{3}}, \frac{q}{\sqrt{3}}\right)$.

$r = 2q^2 \cdot \frac{q^2}{3} - 12 \frac{q^2}{3} \frac{q^2}{3} - 2\frac{q^4}{9} = -\frac{8q^4}{9}$

$s = \frac{4q^4}{3} - \frac{8q^4}{9} - \frac{8q^4}{9} = -\frac{4q^4}{9}$

$t = \frac{2q^4}{3} - \frac{2q^4}{9} - \frac{12q^4}{9} = -\frac{8q^4}{9}$

and $rt - s^2 = \frac{64}{81}q^4 - \frac{16q^4}{81} = \frac{48q^4}{81} > 0$

Since $rt - s^2 > 0$ and $r < 0$ at $x = \frac{q}{\sqrt{3}}, y = \frac{q}{\sqrt{3}}$
$z = \frac{q}{\sqrt{3}}$

i.e. $f(x,y) + v^2 =$ is maximum at $x = y = z$

and hence v is maximum when $x = y = z$

$\Rightarrow$ rectangle solide is cube.

$\therefore$ maximum value is

$V = xyz$

$= \frac{a}{\sqrt{3}} \frac{q}{\sqrt{3}} \frac{q}{\sqrt{3}}$

$V_{max} = \frac{a^3}{3\sqrt{3}}$

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Ex: A rectangular box open at the top is to have a volume 108 cubic. meters. Find the dimensions of the box if its total surface area is minimum.

Soln: ⇒ Let x, y, z be the dimensions of a rectangular box

and let V be the volume and S be the surface area of a rectangular box

i.e $V = xyz$

and $S = xy + 2yz + 2zx$ ——— (open on the upper top)

Let $f(x,y,z) = S = xy + 2yz + 2zx$

put $z = \frac{V}{xy}$

$\Rightarrow f(x,y) = S = xy + 2\frac{V}{x} + 2\frac{V}{y}$ ——— (1)

V is const. $= 108$ cubic

$\Rightarrow \frac{\partial f}{\partial x} = y - \frac{2V}{x^2}$

$\frac{\partial f}{\partial y} = x - \frac{2V}{y^2}$

also, $r = \frac{\partial^2 f}{\partial x^2} = 4\frac{V}{x^3}$

$s = \frac{\partial^2 f}{\partial x \partial y} = 1$

$t = \frac{\partial^2 f}{\partial y^2} = \frac{4V}{y^3}$

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For maximum and minimum value (i.e. extrema values) of $f(x,y)$, necessary condn's are

$$\boxed{\frac{\partial f}{\partial x}=0, \quad \frac{\partial f}{\partial y}=0}$$

ie $$y-\frac{2V}{x^2}=0 \Rightarrow y=\frac{2V}{x^2} \quad \text{---(2)}$$

& $$x-\frac{2V}{y^2}=0 \Rightarrow x=\frac{2V}{y^2} \quad \text{---(3)}$$

Solving (2) and (3) we get.

$$(2) \Rightarrow xy=\frac{2V}{x}=2V\cdot\left(\frac{y^2}{2V}\right)^2=\frac{y^4}{(2V)}$$

$$\Rightarrow y^3=(2V)\Rightarrow \boxed{y=(2V)^{1/3}}$$

$$\& (3) \Rightarrow x=\frac{2V}{y^2}=2V\cdot\left(\frac{1}{(2V)^{1/3}}\right)=(2V)^{1/3}$$

$$\boxed{x=(2V)^{1/3}}$$

Thus the stationary point is $((2V)^{1/3},(2V)^{1/3}) \Rightarrow x=y$

at this point,

$$r=\frac{2V}{x^3}=\frac{2V}{((2V)^{1/3})^3}=2>0$$

$$s=1$$

$$t=\frac{4V}{y^3}=\frac{4V}{((2V)^{1/3})^3}=2$$

and $$rt-s^2=(2)\cdot(2)-1^2=3>0$$

Since $rt-s^2>0$ and $r>0$,

$\Rightarrow f(x,y)=S$, is minimum at $x=y=(2V)^{1/3}$

Hence surface area S is minimum at $((2V)^{1/3},(2V)^{1/3})$

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But $V = 108\ m^3$

$\Rightarrow$ $x = y = (2V)^{1/3}$

$= [2(108)]^{1/3}$

$= (216)^{1/3}$

$= 6$

$\Rightarrow$ $x = y = 6$

and $z = \frac{V}{xy} = \frac{108}{6 \times 6} = \frac{108}{36} = 3$

Hence the dimension of the box which make its total surface area a minimum are

$x = 6,\ y = 6,\ z = 3$

Ex: Divide 24 into three parts such that the continued product of the first, square of second and cube of third is maximum.

Soln: $\Rightarrow$ Let 24 is divide in three part x, y, z

i.e. $x + y + z = 24$ ——— (1)

Now, continued product say $P = x y^2 z^3$

$\Rightarrow$ $f(x, y, z) = x y^2 z^3$

putting ~~$z = 24 - x - y$~~

$(x = 24 - y - z)$ we get

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$$f(y,z) = (24-y-z)\,y^2z^3$$

$$f(y,z) = 24y^2z^3 - y^3z^3 - y^2z^4 \quad \text{——(2)}$$

$$\Rightarrow \frac{\partial f}{\partial y} = 48yz^3 - 3y^2z^3 - 2yz^4$$

$$\frac{\partial f}{\partial z} = 72y^2z^2 - 3y^3z^2 - 4y^2z^3$$

also $r = \frac{\partial^2 f}{\partial y^2} = 48z^3 - 6yz^3 - 2z^4$

$$s = \frac{\partial^2 f}{\partial y \partial z} = 144yz^2 - 9y^2z^2 - 8yz^3$$

$$t = \frac{\partial^2 f}{\partial z^2} = 144y^2z - 6y^3z - 12y^2z^2$$

Now. for stationary point we must have

$$\frac{\partial f}{\partial y} = 0, \text{ and } \frac{\partial f}{\partial z} = 0$$

$$\Rightarrow \frac{\partial f}{\partial y} = 48yz^3 - 3y^2z^3 - 2yz^4 = 0$$

$$\Rightarrow yz^3(48 - 3y - 2z) = 0$$

$$\Rightarrow 48 - 3y - 2z = 0 \quad \text{——(3)}$$

and $$\frac{\partial f}{\partial z} = 72y^2z^2 - 3y^3z^2 - 4y^2z^3 = 0$$

$$\Rightarrow y^2z^2(72 - 3y - 4z) = 0$$

$$\Rightarrow (72 - 3y - 4z) = 0 \quad \text{——(4)}$$

Solving eqn (3) and (4) we get.

$$y = 8, \quad z = 12$$

$\therefore$ (1) $\Rightarrow$ $x = 24 - x - y = 24 - 8 - 12 = 4$

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Hence stationary pt. is (4, 8, 12)

at this point

$$r = 48z^3 - 6yz^3 - 2z^4$$

$$= 48(12)^3 - 6(8)(12)^3 - 2(12)^4$$

$$= -41472 \quad < 0$$

$$s = 144yz^2 - 9y^2z^2 - 8yz^3$$

$$= 144(8)(12)^2 - 9(8)^2(12)^2 - 8(8)(12)^3$$

$$= -27648$$

$$t = 144(8)^2(12) - 6(8)^3(12) - 12(8)^2(12)^2$$

$$= -36864 < 0$$

$\therefore rt - s^2 > 0$

Since at (4, 8, 12) $r < 0$ and $rt - s^2 > 0$

The function $f(x, y, z)$ is maximum

i.e. continued product of first, square of second and cube of third i.e $p = f(x, y, z)$ is maximum at (4, 8, 12)

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Ex: The Temperature T at any point (x,y,z) in a space $T = 400xyz^2$. Find the heighest temp. on the surface $x^2+y^2+z^2=1$.

Soln: let (x,y,z) be the any point on the surface $x^2+y^2+z^2=1$

$$\Rightarrow z^2 = 1-x^2-y^2 \quad \text{---------- (1)}$$

Now let T be the temperature at any pt (x,y,z) is given by $T = 400xyz^2$

$$\Rightarrow f(x,y,z) = 400xyz^2$$

$$\Rightarrow T = f(x,y) = 400xy(1-x^2-y^2) \quad (\because \text{by } (1))$$

$$= 400(xy - x^3y - xy^3) \quad \text{---------- (2)}$$

$$\Rightarrow \frac{\partial f}{\partial x} = 400(y - 3x^2y - y^3).$$

$$\frac{\partial f}{\partial y} = 400(x - x^3 - 3xy^2)$$

also, $r = \frac{\partial^2 f}{\partial x^2} = 400(-6xy) = -2400xy$

$$s = \frac{\partial^2 f}{\partial x \partial y} = 400(1 - 3x^2 - 3y^2)$$

$$t = \frac{\partial^2 f}{\partial y^2} = 400(-6xy) = -2400xy$$

Now, for extreme value we must have

$$\frac{\partial f}{\partial x} = 0 \text{ and } \frac{\partial f}{\partial y} = 0$$

i.e.

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$\frac{\partial f}{\partial x} = 400(y - 3x^2y^2 - y^3) = 0$

$\Rightarrow 400y(1 - 3x^2 - y^2) = 0$

$\Rightarrow 1 - 3x^2 - y^2 = 0$ ——— (3)

and $\frac{\partial f}{\partial y} = 400(x - x^3 - 3xy^2) = 0$

$\Rightarrow 400x(1 - 3y^2 - x^2) = 0$

$\Rightarrow 1 - 3y^2 - x^2 = 0$ ——— (4)

Solving eqn (3) and (4) we get

(3)-(4) $\Rightarrow (1 - 3x^2 - y^2) - (1 - 3y^2 - x^2) = 0$

$\Rightarrow -2x^2 + 2y^2 = 0$

$\Rightarrow -x^2 + y^2 = 0$

$\Rightarrow -x^2 = -y^2 \Rightarrow x^2 = y^2$

$\Rightarrow \boxed{x = \pm y}$

i.e $\boxed{x = y}$ or $\boxed{x = -y}$

For $x = y$: (3) $\Rightarrow 1 - 3x^2 - x^2 = 0$

$\Rightarrow 1 - 4x^2 = 0$

$\Rightarrow x^2 = \frac{1}{4} \Rightarrow \boxed{x = \pm \frac{1}{2}}$

Then for $\boxed{y = \pm \frac{1}{2}}$

For $x = -y$ (3) $1 - 3(-y)^2 - y^2 = 0$

$\Rightarrow 1 - 4y^2 = 0$

$\Rightarrow y^2 = \frac{1}{4} \Rightarrow y = \pm \frac{1}{2}$

$1 - 3x^2 - (-x)^2 = 0$

$\Rightarrow 1 - 3x^2 - x^2 = 0$

$\Rightarrow 1 - 4x^2 = 0$

$x^2 = 1/4$

$\Rightarrow x = \pm$

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$\Rightarrow \boxed{y = \pm 1/2}$

$x = -y$ $\quad \boxed{x = \mp 1/2}$

and Hence probable points are $(\frac{1}{2}, \frac{1}{2})$ $(\frac{-1}{2}, \frac{-1}{2})$, $(1/2, \frac{-1}{2})$ $(\frac{-1}{2}, 1/2)$

Now

	$(1/2, 1/2)$	$(-1/2, -1/2)$	$(1/2, -1/2)$	$(-1/2, 1/2)$
r	-600	-600	600	600
s	-200	-200	-200	-200
t	-600	-600	600	600
$rt-s^2$	360000	360000	360000	360000

Since at $(1/2, 1/2)$, $r < 0$, and $rt - s^2 > 0$

$T = f(x, y)$ has maximum value.

$\therefore T = 400\, xy(1 - x^2 - y^2) = 400 \left(\frac{1}{2}\cdot\frac{1}{2}\right)\left(1 - \frac{1}{4} - \frac{1}{4}\right)$

$= 50^\circ C$

$\boxed{T_{max} = 50^\circ C}$

at $(-1/2, -1/2)$, $r < 0$, $rt - s^2 > 0$

T is maximum at $(-1/2, -1/2) = 50^\circ C$

at $(-1/2, 1/2)$ $\quad r > 0$, $rt - s^2 > 0$

T is minimum $= -50^\circ C$

and at $(1/2, -1/2)$ $\quad r > 0$ $rt - s^2 > 0$

T is minimum $= -50^\circ C$

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